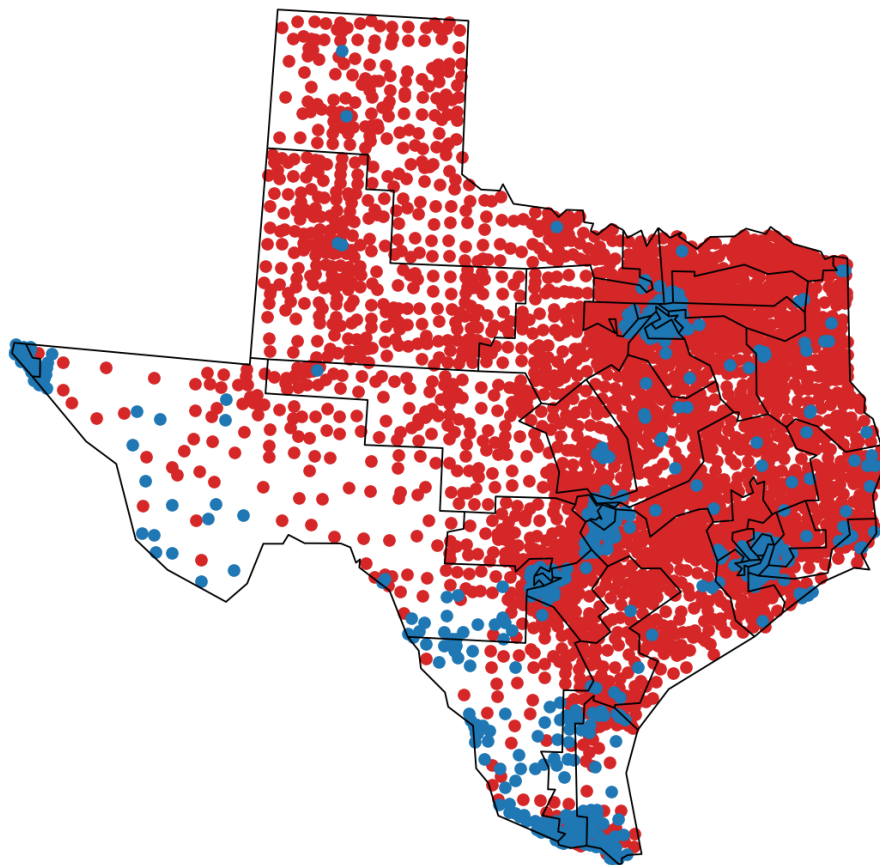


Mapping Democracy

Quantifying Ethnolinguistic Gerrymandering Through the Lens of the Modifiable Areal Unit Problem

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Foreword

I want to express my gratitude to my supervisor, Jennifer, for her unwavering support. I started this thesis with far greater confidence than certainly exceeded my experience. Thanks to her mentorship, I learned that research is a process of continuous reflection. I am grateful for all the support that I received because it helped me to realize that I want to become a researcher as well.

Abstract

Ethnolinguistic gerrymandering influences the representation of ethnic and language minority voters in the biennial elections for the U.S. House of Representatives. While partisan bias in gerrymandering has been widely studied, the U.S. Supreme Court raised the evidentiary bar for challenges to gerrymandering on 23 May 2024 by requiring plaintiffs to provide quantitative evidence of discriminatory intent based on the ethnic or language background of the voters. Partisan bias metrics do not meet this standard because they do not reflect demographic specificity and neglect the modifiable areal unit problem (MAUP). To identify suitable metrics, this study conducted a systematic literature review and compared currently available metrics for two states, namely Texas and South Carolina. Subsequently, this study applied an outlier analysis approach to compare the areal unit segregation (AUS) measure, the geography and election outcome (GEO) metric, and three partisan symmetry metrics. This study found that partisan symmetry metrics alone cannot quantify ethnolinguistic gerrymandering, while AUS reveals district-level patterns of how ethnic and language minority voters are concentrated. The comparison between Texas and South Carolina shows that the interpretability of AUS is sensitive to aggregation and statutory district counts per state, which highlights how MAUP constrains inference differently between states. When used jointly, AUS and GEO remain interpretable under the constraints of MAUP and provide complementary, quantitative evidence to support inference of ethnolinguistic bias. This study demonstrates that no single metric is sufficient. Instead, it argues for a multi-metric, MAUP-aware approach that uses outlier analysis to compare observed spatial patterns to simulated neutral benchmarks, while acknowledging the limits of quantitative inference when applied to ethnolinguistic gerrymandering.

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Abbreviations

Abbreviation	Full Term
AUS	Areal Unit Segregation Measure
EG	Efficiency Gap
GIS	Geographic Information Systems
MAUP	Modifiable Areal Unit Problem
MMD	Mean–Median Difference
MCMC	Markov Chain Monte Carlo
SC	South Carolina
SCOTUS	Supreme Court of the United States
TX	Texas
U.S.	United States
VRA	Voting Rights Act
VTD	Voting Tabulation District

1. Introduction

1.1 Why Ethnolinguistic Gerrymandering Matters

“Redistricting is like an election in reverse. It’s a great event. Usually, the voters get to pick politicians. In redistricting, the politicians get to pick the voters.”

- Thomas Hofeller, Redistricting Consultant (NPR, 2019)

The redrawing of United States’ Congressional District borders, also known as redistricting, is mandated by law to occur after every decennial census to create districts of equal population, so voters are represented fairly in the elections for the U.S. House of Representatives (Figure 1; Bullock, 2010). In these elections, the numbers of seats that a party receives in the House is determined by the total number of votes that it receives, which is dependent on how these votes are geographically distributed and aggregated in electoral districts (Barton & Eguia, 2024). However, the redistricting process may be exploited by political parties to create plans with the intent to favor their own party in the elections, a practice known as gerrymandering (McCartan, 2024). While gerrymandering is not unique to the U.S., the American electoral system is uniquely predisposed to it as neutral commissions are rarely employed in redistricting. Instead, the process is controlled by legislatures where the state’s majority political party holds influence, which opens the door for gerrymandering to occur (Haas, Hachadoorian, Kimbrough, Miller, & Murphy, 2020). Echoing the statement of Thomas Hofeller (NPR, 2019), gerrymandering takes redistricting to its most extreme as voters are strategically denied or permitted to cast a meaningful vote in the elections based on how the borders of districts are drawn. By distorting the intended purpose of redistricting, gerrymandering undermines representative democracy and the fair representation of voters in the U.S. (Yoshinaka & Murphy, 2010).

Gerrymandering takes two distinct forms in the U.S: First, partisan gerrymandering where historical voting outcomes are used to determine the concentration or dilution of a voter bloc across districts which is achieved using two techniques: First, packing which concentrates a voting bloc in a district. Second, cracking which disperses it across several districts to weaken its electoral representation (Stephanopoulos & McGhee, 2014; Zhu, 2024). The 2018 Wisconsin election shows how impactful these techniques can be as Republican candidates secured 63 out of 99 seats in the State Assembly while receiving only 45% of the vote statewide (Palmer, Schnee, & DeLuca, 2023). Second, ethnolinguistic gerrymandering, where data on voter ethnicity and language are used as proxy for voting outcomes and to create favorable districting plans. In essence, the ethnic and language composition of a district is explicitly used to introduce bias, henceforth called ethnolinguistic bias, in the redistricting process with the intent to favor a political party (Canon, 2022). In theory, African Americans historically tend to vote overwhelmingly for the Democrat party across the nation, and an African American-majority district is likely to result in a victory for the Democratic Party (Dawson, 2001; Fairdosi & Rogowski, 2015). In practice, ethnolinguistic gerrymandering is exemplified in the 2012 election for Virginia’s 3rd Congressional district where African American voters constituted 53% of the district’s population. The concentration of a voter bloc in one district, known as packing, contributed to the Democratic incumbent’s victory at 81.3% share of the vote while improving the Republican Party’s chances of victory in neighboring districts (Page v. Virginia State Board of Elections, 2014).

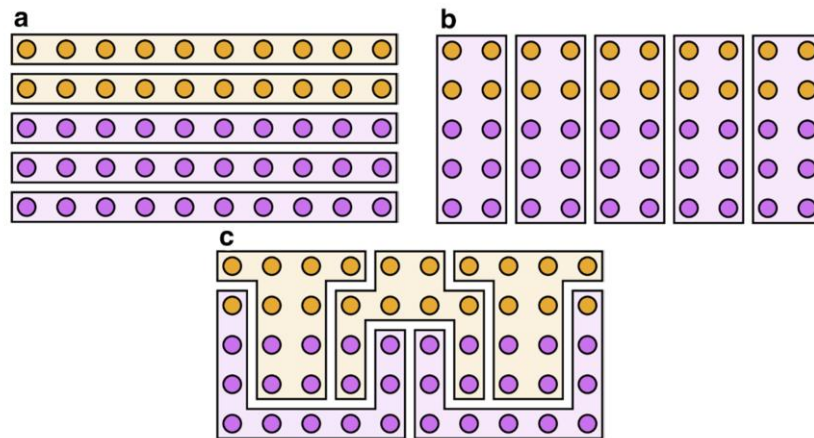


Figure 1. Schematic illustration of how redistricting changes electoral outcomes. Yellow dots represent a voter for party A and purple dots are voters for party B. (a) Composition in which party A wins two districts, B two. (b) Composition in which party B wins all districts. (c) Composition in which party A wins three districts, B two. Source: Guest et al. (2019).

While the issue of gerrymandering and its influence on the representation of voters have not been solved, the practice has been challenged in courts with varying degrees of success. Partisan gerrymandering is legally ambiguous due the ruling of the Supreme Court of the United States (SCOTUS). On one hand, partisan gerrymandering was determined as incompatible with democratic principles. On the other hand, it was also stated to be “beyond the reach of federal courts” because there was no constitutional standard for evidence to show that a constitutional line is infringed upon (*Rucho v. Common Cause*, 2019, p. 6). In essence, partisan gerrymandering is not reviewable and lacks a clear pathway for being challenged in federal courts (Eguia, 2022).

However, ethnolinguistic gerrymandering was explicitly distinguished from partisan gerrymandering by SCOTUS as having a clear constitutional standard for illegality under the Voting Rights Act and as being subject to review by the federal courts (United States, 1965; *Rucho v. Common Cause*, 2019). Citing the VRA specifically, this distinction was further reinforced in *Alexander v. South Carolina State Conference of the NAACP* (2024) where SCOTUS reversed a lower court ruling that successfully challenged a case of gerrymandering. The reversal was under the argument that court rulings are only constitutionally valid if plaintiffs disentangle ethnolinguistic from partisan gerrymandering and isolate the intent to discriminate against ethnic and language minorities listed in the VRA. These minority voter groups are defined as African Americans, Hispanic Americans, Asian Americans and Native Americans (United States, 1965). For challenges to gerrymandering to be successful, plaintiffs must show that the districting plan was in violation of the VRA by providing empirical evidence that ethnicity and language were the predominant factors that determined the redistricting outcome (*Alexander v. South Carolina State Conference of the NAACP*, 2024). In essence, there must be evidence of discriminatory intent to show that gerrymandering occurred on illegal ethnolinguistic grounds, rather than the partisan grounds which fall outside of the federal court’s purview. Practically, this evidence is difficult to provide for the judiciary as the two types of gerrymandering cannot be easily disentangled and intent is difficult to express in quantitative, empirical evidence (Ramachandran & Gold, 2018; Ross, 2024).

1.2 How to Understand Gerrymandering with GIS

While the distinction between partisan and ethnolinguistic gerrymandering is blurry, geographic information systems (GIS) offer a computational approach to quantify gerrymandering. GIS software and its optimization tools are invaluable for practitioners of gerrymandering (Plane, Tong, & Lei, 2018). By displaying and manipulating massive quantities of demographic, geographic and electoral outcome data, GIS makes it possible to generate and analyze different district configurations with the goal of identifying the optimal one (Monmonier, 2001). Because GIS is used as a tool in gerrymandering practices, it can also be used to challenge them. However,

quantitative tools cannot identify discriminatory intent unless they are combined with a suitable theoretical framework to analyze and interpret the outcomes.

The foundation of this framework lies in the theory of the Modifiable Areal Unit Problem (MAUP). MAUP refers to the variation in the results of aggregating smaller units of data into larger ones, and how these outcomes are interpreted (Openshaw, 1983). For example, when point-based data is aggregated into administrative units such as blocks or the larger precincts, the outcome of the aggregation in ethnic composition can vary strongly depending on how the borders of the units were drawn. Redistricting, and thus gerrymandering, exemplifies the MAUP because it involves demographic data at bloc precinct level, which is aggregated from points, being further aggregated into congressional districts. Blocs, precincts and districts are all modifiable areal units created through aggregation based on administrative boundaries. Any modification of these boundaries may alter the demographic profile of the district and lead to different electoral outcomes, which means that gerrymandering inherently involves MAUP (Guest et al., 2019; Powell, 2021).

The use of GIS to study gerrymandering through the lens of the MAUP has an extensive body of literature. Shirabe (2002) used GIS to systematically address issues arising from MAUP by optimizing the allocation of spatial units. This MAUP-focused approach is applied to gerrymandering specifically by Oehrlein and Haunert (2017) who showcase that integer linear programming and constraint-based aggregation offer precise and transparent ways to address the aggregation and contiguity problems of MAUP. Furthermore, Saporito and Maliniak (2022) apply a GIS-based approach to systematically study and distinguish gerrymandered districts from those where the electoral representation of minority voters matches that of its neighboring districts using the areal unit segregation measure. As such, studying gerrymandering as a case of MAUP with a GIS-based approach opens new possibilities in terms of insights which can be gained regarding evidence of intent to discriminate against ethnic and language minority voters as outlined in the VRA and ruled in *Alexander v. South Carolina State Conference of the NAACP* (2024).

1.3 The Challenge of Quantifying Gerrymandering

Despite extensive academic and legal discourse, there is no method that systematically quantifies the effects of gerrymandering on the electoral representation of ethnic and language minority voter groups in the biennial elections for the U.S. House of Representatives. Existing assessment methods and their metrics can offer insights into the partisan bias or geographic characteristics of districting plans. However, they do not quantify ethnic and linguistic bias in isolation on a national scale to meet the standard for evidence admissible to federal courts as established in *Alexander v. South Carolina State Conference of the NAACP* (2024), nor are they suitable for providing evidence of discriminatory intent in judicial cases challenging gerrymandering. The challenge of developing such a method is complicated by the lack of an objective standard of what constitutes clear empirical evidence of discriminatory intent, or of what constitutes fair electoral representation of a specific voter bloc (Haas, Hachadoorian, Kimbrough, Miller, & Murphy, 2020).

This study seeks to describe the current state of ethnolinguistic gerrymandering's effects on minority voter representation through the geographical lens. By using the MAUP to contextualize the study's findings, it is possible to understand which of its conclusions are attributable to MAUP rather than the ground truth. The precedent set by SCOTUS highlights the need for a GIS-based to systematically quantify the current effects of ethnolinguistic gerrymandering on a national scale in the U.S and to decouple it from metrics of partisan bias. In this study, the biennial elections for the U.S. House of Representatives are chosen specifically due to their importance at the national scale in prominent court cases challenging gerrymandering. Additionally, the frequency of these elections leads to a longer time series where the assessment method can be applied. By building on existing assessment methods, this study aims to advance academic understanding of how

gerrymandering affects electoral representation and to inform the legal and policy debates around gerrymandering cases in the U.S.

1.4 Research Objective & Questions

The objective of this research is to provide quantitative insights into how ethnolinguistic bias can be inferred from the spatial patterns of gerrymandering, while accounting for the constraints imposed by MAUP. These insights are created by identifying an assessment approach comparing possible metrics to systematically study ethnolinguistic gerrymandering in biennial U.S. House of Representatives elections. The main research question is as follows:

What is the role of spatial patterns of gerrymandering in supporting the inference of bias against ethnic and language minority voters in the biennial U.S. House of Representatives elections under constraints imposed by the modifiable areal unit problem?

The sub-research questions are as follows:

1. What assessment methods and metrics are currently used and what are their limitations?
2. Which are relevant to quantify the spatial patterns of ethnolinguistic gerrymandering?
3. What modifications are needed to quantify the spatial patterns of ethnolinguistic gerrymandering?

1.5 Structure of the Report

The report is structured as follows: Chapter 2 describes MAUP in detail as the theoretical framework of this study. Chapter 3 contains the materials and methods, and it explains the use of PRISMA guidelines to conduct a systematic literature review. Additionally, the chapter describes how spatial data was collected and analyzed according to the outlier analysis method chosen to quantify gerrymandering. The outlier analysis method is described in detail in addition to the GEO and the areal unit segregation metrics as applied in this study. Chapter 4 gives the results of the study with literature review in 4.1 as a starting point where the identified methods and metrics used to quantify gerrymandering are listed. Subchapter 4.2 gives the results of the spatial data analysis for the 2020 House election in South Carolina and 2026 in Texas. Chapter 5 contains the discussion where the results of outlier analysis as applied in this study are placed in the context of scientific literature on geographic approaches to quantify gerrymandering. The limitations of this study and recommendations for further research are given as well. Chapter 6 concludes the report by answering the main research question.

2. Theoretical Framework

2.1 Gerrymandering as the Modifiable Areal Unit Problem

Gerrymandering can be viewed as an aggregation problem where the combination of smaller spatial units of data into larger units alters statistical relationships and the interpretation of the results. As stated in the Introduction, redistricting is inherently linked to the Modifiable Areal Unit Problem (MAUP). MAUP was first defined by Openshaw (1984) as a geographic problem caused by large number of ways in which continuous data can be arbitrarily divided into areal units with different results and interpretation as a consequence. The framework consists of two sub-problems: First, the scale problem where the aggregation of smaller scale data points such as blocks and census tracts into electoral districts leads to different results with increasing scales. Second, the aggregation problem where variation occurs in the results of analysis due to the use of alternative units. MAUP not only demonstrates how aggregation affects the results, but also that every choice in spatial aggregation reflects both conscious and sub-conscious bias of the aggregator. The identification of gerrymandering as an example of MAUP is not novel as it was recognized in the field of geography before the concept was defined.

Before MAUP was defined in Openshaw (1984), Openshaw (1977) first demonstrated empirically that statistical outcomes from spatial models were affected by the chosen scale and aggregation. Additionally, it identified that variations in zoning systems and spatial aggregation affect model parameters, the aggregation problem. This effect of aggregation means that zoning choices significantly impact the interpretation and performance of spatial models. Expanding on the aggregation problem, Openshaw (1978) proposed that optimized zoning systems can mitigate the MAUP by reducing variability in the quantitative outcomes of models, with the argument that zone design must be carefully considered in spatial modelling. Openshaw (1984) defined the MAUP and placed it in the context of several discussions within the domain of geography, such as identifying gerrymandering as a case of MAUP where zone design seems to not be carefully considered at all. Masser and Brown (1978) also identified that spatial interaction analyses were markedly influenced by zoning decisions, further recommending that intention and information are what drives spatial aggregation, and it may exacerbate or mitigate the zoning problem. Building upon these foundations, Fotheringham and Wong (1991) expanded further on MAUP's implications by demonstrating that its effects in multivariate statistical analyses are highly unpredictable and significant, which led to the recommendation that MAUP should be addressed. They found that changes in spatial scale and zoning schemes resulted in significant variations in parameter estimates and model interpretations, often leading to contradictory conclusions. By identifying MAUP's effects in multivariate statistical analyses, its role in electoral geography not only became clearer, but it became possible to quantify the exact effects of MAUP within geography further.

Lee, Rogers, and Soifer (2024) place the frameworks established by Openshaw (1984) and Fotheringham and Wong (1991) firmly within the domain of electoral geography. They demonstrate that both dimensions of the MAUP, the scale and aggregation problems, influence statistical relationships outcomes of electoral studies. For example, interpretation of electoral outcomes, voting patterns, and representation metrics are highly sensitive to the aggregation of demographic and voting data into districts. This means that the MAUP also determines the electoral outcomes and representation in cases of redistricting and ethnolinguistic gerrymandering.

2.2 Population Thresholds in Redistricting and Representation

Electoral representation entails the process by which citizens' preferences are translated into legislative seats, which makes it a central concept to understanding the effects of ethnolinguistic gerrymandering. Grofman, Handley, and Niemi (1992, p. 133) define electoral representation as acting in the interest of the represented, in a manner responsive to them. For representation to be fair, it must “*require that numbers of representatives bear some relationship to the kinds of individuals represented*”. Within the context of gerrymandering, this explicitly means that an electoral district should enable minority voter groups to cast votes that meaningfully influence election outcomes rather than become “wasted” votes. Following Stephanopoulos and McGhee (2014), wasted votes include those cast either for losing candidates or in excess of the number necessary for a winning candidate to secure victory. Therefore, fair electoral representation of minority voter groups is achieved when electoral district boundaries minimize the number of wasted votes, allowing these groups to affect the outcome of elections.

Fair electoral representation can be quantified, and it involves systematically evaluating whether minority voter groups can effectively elect their preferred candidates. Grofman, Handley, and Niemi (1992) propose empirical criteria to assess minority representation, emphasizing measurable outcomes such as the number of elected minority representatives and the extent to which election results reflect minority voter preferences. Furthermore, they formulate approaches more suitable for a spatial-analysis approach which includes analyzing voting patterns to determine the minimum percentage of minority voters required within a district to achieve meaningful representation outcomes. This method, threshold of representation analysis, explicitly quantifies the relationship between minority population proportions within districts and their electoral success. Handley (2020) expands on the threshold of representation by emphasizing structural and systemic factors in congressional redistricting. They show that by exploring how deliberate district design, such as creating majority-minority districts, where minority voters outnumber the national majority, can enhance the ability of minority voters to effectively affect electoral outcomes. This however may also have the effect of reducing the electoral representation of minority voters in the surrounding district as they are effectively packed within just one (Stephanopoulos and McGhee, 2014). The reliance on precise metrics of minority populations combined with the analysis of a district’s exact design makes the framework established by Grofman, Handley, and Niemi (1992), and Handley (2020) particularly suitable for examining ethnolinguistic gerrymandering through the lens of the MAUP.

2.3 Ethnolinguistic Gerrymandering and Population Thresholds

The minimum threshold of minority voters is a natural part of MAUP as underscores that the spatial configuration and aggregation choices determine the electoral outcomes and thus the representation of minority voter groups. Grofman, Handley, and Niemi’s (1992) threshold of representation analysis is relevant within the MAUP framework because it highlights spatial scales and districting schemes to ensure minority voter groups' votes translate accurately into fair representation outcomes. Recognizing and addressing MAUP in electoral district design is therefore critical for maintaining fair electoral representation of minority voter groups.

3. Materials and Methods

3.1 Study Areas

3.1.1 South Carolina: 2022 House Elections

South Carolina's relevance as a recent high-profile case in ethnolinguistic gerrymandering stems from its role in *Alexander v. South Carolina State Conference of the NAACP* (2024). In this court case, a panel of three federal judges initially found the state's 1st District of the enacted plan S. 865 to be gerrymandered in favor of the Republican Party. This decision was later reversed by the U.S. Supreme Court, which established a stricter evidentiary bar for proving ethnolinguistic gerrymandering for all cases moving forward. Before, South Carolina was the focus of five redistricting cases from 1992 to 2012. For example, *Backus v. South Carolina* (2012) also argued that the post-2010-census redistricting plan violated the VRA by discriminating against ethnic minority voters, but this case was rejected in local courts.

South Carolina is the 8th state of the U.S., and it is in the country's Southeast (Figure 2). It is the 11th-smallest state of the U.S. with a total area of 82,932.7 km², 6.12% of which is water (U.S. Census Bureau, n.d.-a). The state's total population equals 5,478,831 with 150,227 residing in Charleston, South Carolina's largest city, which is located on the state's coast in the Southeast. The Charleston metropolitan area is estimated to have 869,940 residents. The second largest urban agglomeration by population is South Carolina's capital, Columbia, at 136,632 residents with its metropolitan area estimated at 856,889 residents. Columbia is located in the center of the state (U.S. Census Bureau, 2021). In the ascending order of administrative scales, South Carolina has: 146,844 census blocks, 2,351 precincts, 2,268 voting tabulation districts, 46 counties, and seven congressional districts (U.S. Census Bureau, Geography Division, 2020-a). In 2022, District 1 included most of Charleston, while District 6 contained Columbia and the northern part of Charleston. Other districts consisted primarily of rural areas except for District 5, which also included the suburbia of Charlotte, North Carolina (South Carolina Senate Judiciary Committee, 2022).

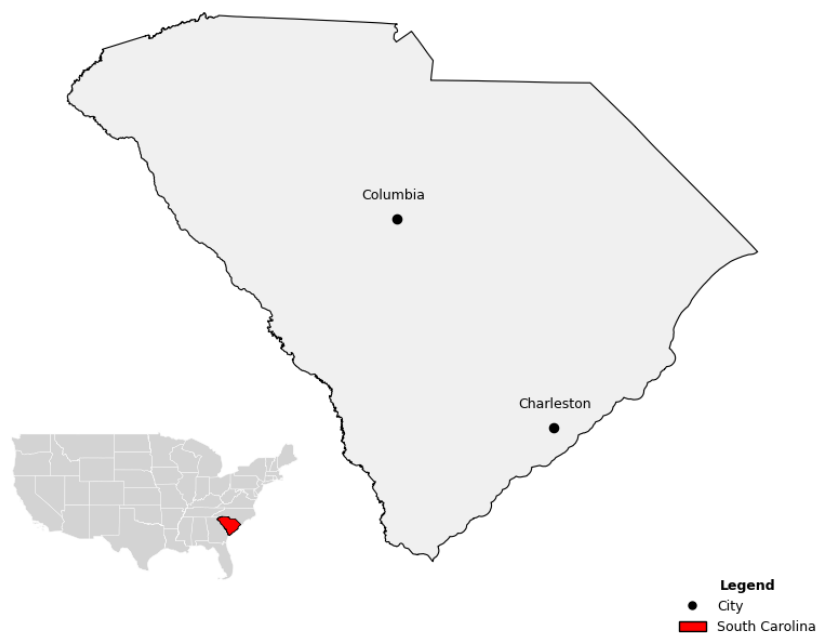


Figure 2. Relative location of South Carolina in the U.S. and its main metropolitan hubs.

For the 2022 election, South Carolina enacted plan S.865 which had seven districts total (Figure 3). Six districts were won by the Republican Party, and one, District 6, by the Democratic Party, with 1,056,078 and 517,129 votes, respectively (Federal Election Commission, 2025). At the time of the 2020 census, which was relevant for the 2022 election, the state had a population of 5,118,425 with the following breakdown by ethnic background: 24,303 American Indian and Native, 90,466 Asian, 1,280,531 African American, 352,838 Hispanic, 3,431 Native Hawaiian and Pacific Islander, and 3,243,442 White. In sum, 34.2% or 1,751,569 of the population was an ethnic minority following the definition in the VRA. In terms of language, 8.5 % or 435,066 of the population spoke a different language from English at home (U.S. Census Bureau, 2021).

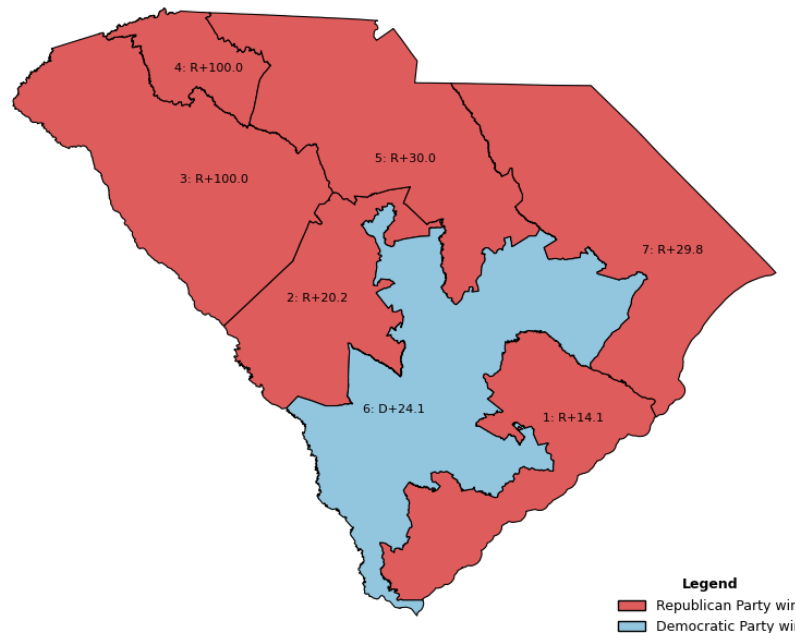


Figure 3. South Carolina 2022 House election districts and returns. The margin of victory for the winning party is given as a label for each of the seven districts. For example “6: D+24.1” means that for District 6, the Democratic Party secured 24 % points more votes than the Republican Party.

3.1.2 Texas: 2026 House Elections

In 2025, Texas became the focus of media discourse on gerrymandering due to the enactment of PLANC2333 which was argued to strongly favor the Republican party (Astudillo, 2025). Historically, Texas had several court cases challenging gerrymandering with numerous taking place between 1973 and 2025. Particularly relevant was *Perry v. Perez* (2012), a Supreme Court case, which set standards for drawing interim Congressional maps. *United States v. Texas* (2021) was a similarly notable case as the Department of Justice sued the state arguing that ethnolinguistic gerrymandering took place in the 2021 enacted plan favoring the Republican Party. However, the case was withdrawn in 2025 following the inauguration of the 47th President of the United States (U.S. Department of Justice, 2021; The American Redistricting Project, 2025).

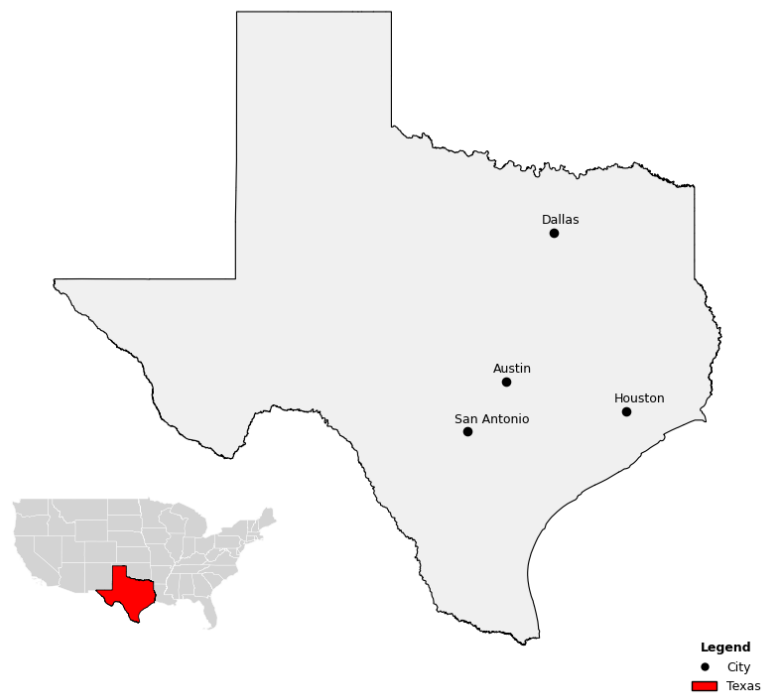


Figure 4. Relative location of Texas in the U.S. and its main metropolitan hubs.

Texas is the 28th state of the U.S., and it is in the country's South (Figure 4). It is the 2nd-largest state of the U.S. with a total area of 695,662 km², 2.7% of which is water (U.S. Census Bureau, n.d.-b). The state's total population equals 31,290,831 with 2,304,580 residing in Houston, the largest city of Texas, which is located on the coast in the southeast of the state. The Houston metropolitan area is estimated to have 7,824,643 residents. The second largest urban agglomeration by population is Dallas in the North of Texas. It has 1,304,379 residents with the Dallas-Fort Worth metroplex estimated at 7,637,387 residents. Austin, Texas's capital, is located centrally in the state with a population of 961,855 (U.S. Census Bureau, 2021). In the ascending order of administrative scales, Texas has: 668,757 census blocks, 9,014 precincts, 9,157 voting tabulation districts, 254 counties, and 38 congressional districts (U.S. Census Bureau, Geography Division, 2020-b).

In the 2020 election, Texas had 38 districts with 25 being won by the Republican Party at 5,926,712 votes and 13 by the Democratic Party at 4,896,383 votes (Johnson, 2021). At the time of the 2020 census, which will be relevant for the 2026 election, Texas had a population of 29,145,505 with the following breakdown by ethnic background: 226,866 American Indian and Native, 1,561,518 Asian, 3,444,712 African American, 11,441,717 Hispanic, 58,291 Native Hawaiian and Pacific Islander, and 11,584,597 White. In sum, 57.4% or 16,733,104 of the population was an ethnic minority following the definition in the VRA. In terms of language, 35.4% or 10,317,508 of the population spoke a different language from English at home (U.S. Census Bureau, 2021).

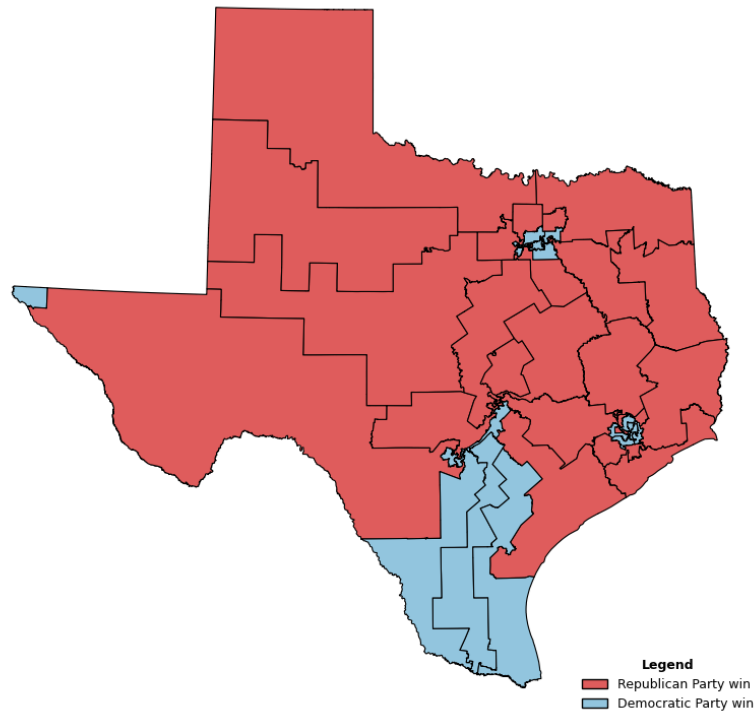


Figure 5. South Carolina 2020 House election districts and returns.

3.2 Systematic Literature Review

In the systematic literature review, I use modified PRISMA guidelines with the objective of identifying and comparing the limitations of the currently used assessment methods that quantify gerrymandering (Figure 6; Page et al., 2021). The PRISMA guidelines are a standardized framework to document and report the systematic literature review process in a transparent and replicable manner (Snyder, 2019).

I used the following query on Web of Science, which was chosen because it is a database that features solely peer-reviewed scientific literature, with the results reported in Figure 3:

TS=((gerrymander* OR redistrict* OR "electoral district*" OR "electoral represent*" OR congress* OR maldistrict*) AND (rac* OR ethn* OR "partisan" OR minorit*))

AND TS=("spatial analysis" OR geospatial OR "geographic information system*" OR GIS OR cartograph* OR mapping OR "spatio-temporal" OR "spatial segregation" OR "areal unit" OR "computational redistrict*" OR simulation OR algorithm* OR "automated redistrict*" OR optimization OR modeling)

AND TS=(quanti* OR causal OR metric* OR index* OR indicator* OR statistic* OR measure* OR procedure* OR reform* OR criteria OR method* OR assess*)

AND TS=("United States" OR America* OR U.S. OR US OR USA OR "American politics")

NOT TS=(covid* OR coronavirus OR pandemic OR medic* OR school* OR student* OR teacher* OR education OR plant* OR crop* OR forest* OR agriculture OR vietnam OR asia OR africa OR china)

AND PY=(2014-2025)

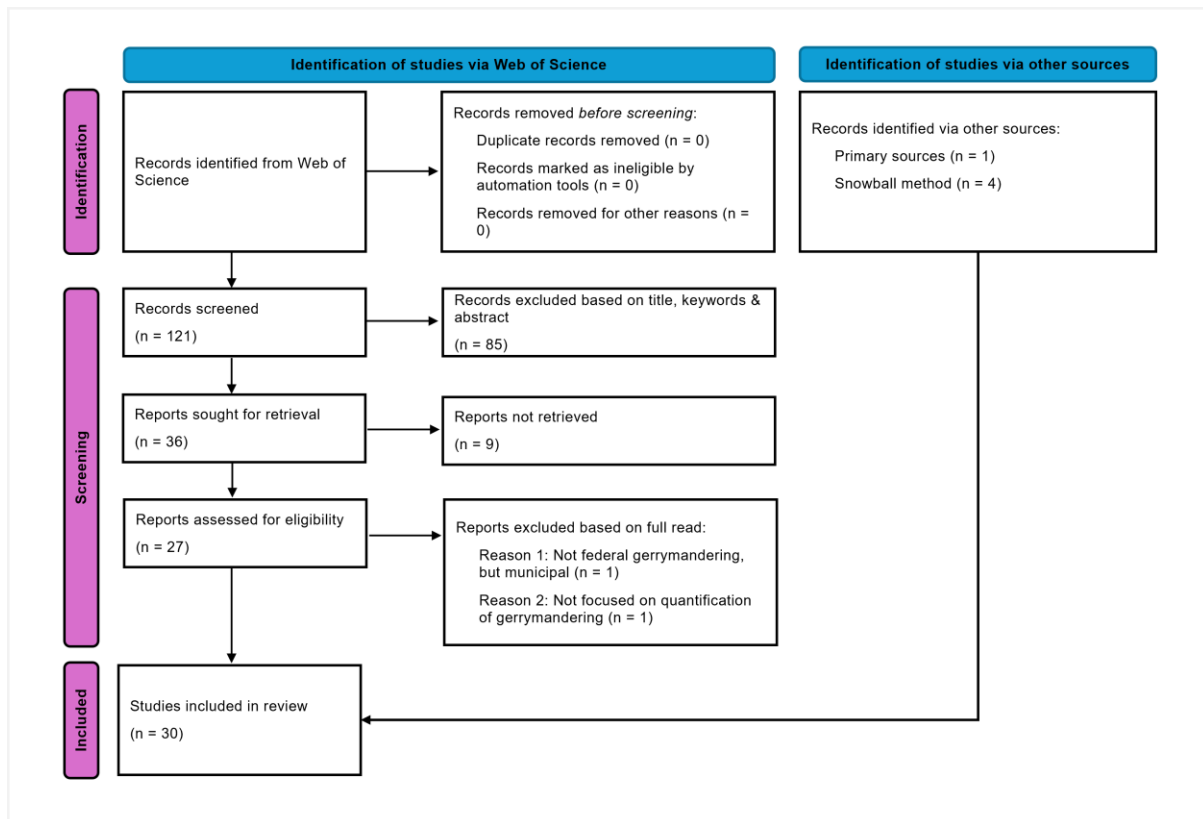


Figure 6 Reporting of results through the four phases of a systematic review. Adapted from Page et al. (2021, p. 5).

After applying the search string, I included peer-reviewed articles if they explicitly address the quantification of gerrymandering through assessment methods and indicators. Articles were included only if they discuss the U.S. and I excluded studies published before 2014, the publication year of Stephanopoulos and McGhee (2014), which I see as a seminal paper. Additionally, I excluded non-ISI-listed and non-English works without official translations. As an exception to the established guidelines, I still considered several papers published before 2014 in the literature review because they define key concepts and methods and were referenced in the collected literature. The complete list of selected literature was included in Annex B.

I analyzed the selected literature using a deductive coding approach to identify the assessment method and its metrics, parameters, data requirements, and limitations. I use this deductive approach because it ensures consistency in comparing findings from literature. Using a deductive coding framework, I coded each selected paper according to the type of bias it assesses, partisan or ethnolinguistic, the assessment method used, the indicator's name, input data required, scale of assessment, and its limitations (Table 5). I defined an assessment method as the general approach or technique used to quantitatively evaluate whether gerrymandering occurs and what its effects are. A metric refers to the quantitative measure used in the assessment method such as the efficiency gap. The parameters are the input values used to calculate the metric such as district-level population data. To support the coding, I used the Atlas.ti software without employing its AI toolset.

Table 1. Overview of coding themes.

Coding Theme	Description	Examples
Type of Bias	Specifies the form of bias targeted by the indicator.	Partisan bias, ethnolinguistic bias
Type of method	Type of assessment method following the categories of Haas, Hachadoorian, Kimbrough, Miller, & Murphy (2020).	Optimization, heuristic
Judicial connection	Whether the assessment method references a specific legal precedent or court case as the primary influence on its development or application.	<i>Page v. Virginia State Board of Elections</i> , 2014; <i>Alexander v. South Carolina State Conference of the NAACP</i> , 2024
Metric	Specific indicator or metric proposed by the authors to quantify gerrymandering.	Efficiency gap, mean-median difference, compactness
Parameters	Variables defined and used by the author to calculate the indicator.	Wasted votes, border length
Data requirements	The type of data needed to apply the method.	District boundaries, demographic data
Spatial coverage	The geographic level at which the analysis was conducted.	Applied only to districts within the state of Florida
Temporal coverage	The timeframe covered in the analysis by the author.	Applied only to 2022 election cycle
Limitations	Methodological, conceptual, or practical limitations of the indicator or method given by the authors.	Assumes uniform voter distribution, unable of distinguishing intent in gerrymandering/redistricting

3.3 Data Requirements and Pre-processing

This study requires information on population and electoral outcomes to simulate potential redistricting plans. Suitable ready-made datasets are not available for the study areas (MGGG, 2024). As such, I create the datasets following the workflow described by DeFord et al. (2023), where datasets available at different administrative scales are aggregated at the level of voting tabulation districts (VTD). A VTD is a geographic unit used by the Census Bureau to compile election data for redistricting, and while it generally follows precinct boundaries, it may differ from them across election cycles. The hierarchy for the administrative scales is as follows: census block, precinct, VTD, electoral district, and state. All used datasets are listed in Annex A.

To construct the South Carolina 2022 House election dataset, I first harmonize the VTD layer by dissolving geometries according to their ID to ensure a single feature with valid geometry per VTD. Second, I aggregate the demographic data by first coercing the total populations in the census

block shapefile to the numeric data type and then areal weight allocated the block population totals into the VTDs. Third, I aggregate the electoral outcomes from the precinct layer by initially calculating the vote totals per precinct for the Republican Party and Democratic Party. I then use areal-weight allocation to move the precinct totals into the VTDs. Fourth, I left-join the aggregated population and vote totals to the VTD geometry and replace residual nulls with zeros. Fifth, I validate the resulting dataset by comparing the total population in the census block dataset with the total population sum for the aggregated VTD dataset. I make the same comparison using the total number of votes for all candidates in the precinct dataset and the aggregated VTD dataset. I repeat the exact same process for the Texas 2026 election but use the precinct-level returns of the 2020 election as proxy for the 2026 data. The 2022 and 2024 results were not used because the 2020 election results were the closest match with the 2020 census data. The 2020 election results could thus serve as proxy for 2026 because the districts in this year are also drawn using the 2020 census.

3.4 Outlier Analysis: General Method

Outlier analysis is a concept taken from statistics which involves the identification of data points with values that deviate significantly from the other data points. In the context of statistics, an outlier is defined as “an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism” (Aggarwal, 2017, p. 1). The outlier is identified through different methods, one being a threshold of 1.5 times the interquartile range of all data points (Saporito & Maliniak, 2022).

This study uses outlier analysis as a way to identify intent by comparing selected metrics calculated for an enacted redistricting plan with a simulated ensemble of valid plans, which serves as a neutral benchmark. The ensemble is neutral when it is produced without data on ethnicity or language (Chen & Rodden, 2015). If an enacted plan is an outlier compared to the values calculated for the neutral ensemble, it can be inferred that data on ethnicity and language was used in the creation of the enacted plan. By identifying the enacted plan as an outlier, quantitative evidence is given of discriminatory intent, which in the context of litigation shifts the burden to the defendant to give legitimate non-discriminatory reasons for the disparity in the values calculated for the enacted and simulated plans (Ramachandran & Gold, 2018; *Common Cause v. Rucho*, 2018).

Outlier analysis follows a general approach as described by Ramchandran & Gold (2018). This approach uses the precinct geometries for an entire state with population totals aggregated from census blocks. Additional data requirements depend on the selected metrics. The analysis starts by defining constraints, namely contiguity, single-object districts, and population equality. For these, districts must have population totals within a tolerance threshold. The constraints are implemented per state using an algorithm with the following steps (Chen and Rodden (2015): (1) Label each precinct as a district. (2) Randomly select one district and choose its closest district neighbor. (3) Combine the selected districts. (4) Repeat until the specified number of districts is achieved for the state. With these four steps, the algorithm ensures that all created districts are contiguous.

To achieve population equality, the algorithm continues (Chen and Rodden, 2015): (5) Select two neighboring districts with highest total population difference. (6) In the higher population district, identify precincts than border the neighbor. (7) Select the precinct in the higher population district which is closest to the neighbor. (8) Transfer the precinct to the lower population district. (9) Repeat until all districts meet the population equality constraint. By following these steps, a neutral redistricting plan is generated that follows the constraints of contiguity and populating equality, making it a valid plan. Afterwards, more plans are generated using the same steps and for each plan, the values of selected metrics are calculated and saved (Ramachandran & Gold, 2018).

The baseline approach described above carries several limitations as identified by Ramchandran & Gold (2018): First, it does not guarantee representative sampling due to its tendency to over- and under-sample parts of the state if the area is small or the number of precincts low. This means that the algorithm may fail to construct a comprehensive, representative benchmark. Second, the algorithm is “ad hoc and lacks a mathematical proof that it adequately samples the map space” (Ramchandran & Gold, 2018, p. 293). Third, the closest-centroid definition used to select precinct and districts lacks sufficient justification and presents the danger that outcomes of the algorithm hinge on authors’ design choices.

While the above is the baseline approach of outlier analysis as established by Chen and Rodden (2015), several alterations exist which build upon this method and address its limitations. One such method that I selected for this study is described further in subchapter 3.6.

3.5 Metrics Used in Outlier Analysis

Outlier analysis facilitates the integration of metrics to quantify gerrymandering. However, not all of these metrics are suitable for quantifying the effect of ethnolinguistic gerrymandering on the representation of minority ethnic and language voter groups. For this study, I selected five metrics with an overview found in Table X.

Table 2. Overview of the five gerrymandering metrics applied in this study.

Metric	Short Description	Judicial Application	Key Limitations	Relevance to This Study
Efficiency Gap	Measures partisan symmetry through differences in wasted votes between parties.	Vieth v. Jubelirer (2004); Gill v. Whitford (2018)	Assumes 50/50 vote split as fair. Has no spatial or ethnolinguistic component.	Useful to identify partisan gerrymandering but insufficient for ethnolinguistic gerrymandering. Included as parallel indicator.
Mean–Median Difference	Assesses skew in district-level vote shares using mean–median comparison.	League of Women Voters v. Pennsylvania (2018)	Not a spatial metric and assumes symmetry implies fairness. Has no spatial or ethnolinguistic component.	Limited added value, but included as parallel indicator.
Partisan Bias	Quantifies deviation from partisan symmetry using seats–votes curve.	LULAC v. Perry (2006); Gill v. Whitford (2018)	Relies on counterfactual assumptions and may give paradoxical outcomes. Has no spatial or ethnolinguistic component.	Useful to identify partisan gerrymandering but insufficient for ethnolinguistic gerrymandering. Included as

				parallel indicator.
GEO Metric	Counts potential competitive districts based on vote share swaps.	Rucho v. Common Cause (2019)	Assumes uniform turnout; may generate implausible district configurations.	Valuable as MAUP-aware metric and included as parallel indicator.
Areal Unit Segregation Measure	Compares demographic district composition to neighboring districts to detect packing/cracking.	Bethune-Hill v. Virginia Board of Elections (2018)	High data requirements; no fixed constitutional threshold.	Core metric for MAUP-aware assessment of ethnolinguistic gerrymandering.

3.5.1 Efficiency Gap

Conceptualized by Stephanopoulos and McGhee (2014), the efficiency gap (EG) quantifies partisan symmetry in electoral outcomes as a result of partisan gerrymandering. On its own, the efficiency gap is a type of vote-dilution assessment method which uses the indicator of wasted votes which are votes cast for a losing candidate or votes in excess of the fifty percent threshold required to win the election (Douchin and Walch, 2017; Eq. 1). If the calculated efficiency gap equals zero, it denotes ideal partisan fairness which indicates that no partisan gerrymandering has occurred. The value thresholds which indicate that unconstitutional partisan gerrymandering has occurred are 2 seats for Congressional elections. To calculate it, district-level election returns, and the total vote counts for each electoral district are required. The efficiency gap assessment method is applicable nationally at the scale of congressional districts with the initial application using a temporal coverage of the years 1972-2021 (Stephanopoulos and McGhee 2014).

$$(1) \quad EG = \frac{Wasted\,Votes_{Party\,A} - Wasted\,Votes_{Party\,B}}{Total\,Votes}$$

The efficiency gap addresses the SCOTUS objection to *Vieth v. Jubelirer* (2004) which argues that there is no judicially manageable standard available to assess partisan gerrymandering. EG addresses this argument by using data only from actual electoral results rather than estimates of swing votes as result of gerrymandering. The threshold of two seats is also a standard argued to be judicially manageable. The method's legislative focus led to its use in the *Whitford v. Gill* (2016) case as quantitative evidence of partisan bias resulting from gerrymandering where plaintiffs successfully demonstrated a pro-Republican bias across multiple election cycles in Wisconsin, but SCOTUS still emphasized that quantitative evidence alone cannot conclusively identify whether gerrymandering has occurred.

The efficiency gap is limited by its assumption that the ideal vote share equals 50% while statewide vote shares are rarely split equally between the Republican and Democratic parties. Additionally, the 2-seat threshold is not derived from existing scientific theory, but rather precedent identified by the authors. Zhu (2024) argues that the threshold is arbitrary and does

not account for state-specific political geographies such as in New York where the threshold is regularly surpassed due to high clustering of Democrat voters. Finally, the method does not acknowledge ethnolinguistic gerrymandering, nor does it include any analysis of spatial factors. Stephanopoulos & McGhee (2018, p. 1510) evaluate it as follows: “*No metric is perfect. But the EG performs as well or better than any other available measure against our evaluative criteria*”. However, they also advise using the efficiency gap integrated as a metric in other assessment methods to address the limitations of the methods.

For quantifying ethnolinguistic gerrymandering, the efficiency gap is useful for clearly identifying whether gerrymandering occurs. However, it requires integration with other methods of assessment as it does not quantify ethnolinguistic gerrymandering nor its effects. Additionally, the efficiency gap lacks spatial components which may make it unsuitable if MAUP is the primary theory behind the assessment method.

3.5.2 Mean-Median Difference

First defined by McDonal and Best (2015), the mean-median difference (MMD) quantifies partisan gerrymandering by measuring the skew in a party’s district-level vote shares. The skew is calculated as the difference between a party’s average vote share across all districts and its median vote share (Eq. 2). If MMD equals 0, no evidence of partisan gerrymandering is found. If it is below zero, it indicates that the party has packed districts while a positive value indicated that cracking occurred.

$$(2) \quad \text{MMD} = \text{Average Vote Share} - \text{Median Vote Share}$$

The logic behind the MMD is that it identifies potential cases of gerrymandering using actual electoral outcomes. It has been used in the case of *League of Women Voters v. Commonwealth of Pennsylvania* (2018) as evidence that the enacted redistricting plan was as outlier compared to other cases of redistricting.

The metric has several limitations such as the assumption that if the mean and median vote shares are equal, there is no evidence of partisan gerrymandering (Zhu, 2024). For example, the mean vote share across Democrat strongholds such as in New York is expected to be higher than the median vote share of the party across districts in other states. Moreover, the MMD method does not incorporate any spatial data.

In terms of relevancy for a new assessment method of ethnolinguistic gerrymandering, MMD may serve as an additional metric. However, no strengths of the method were identified compared to other approaches which quantify partisan gerrymandering.

3.5.3 Partisan Bias

The partisan bias metric was first implemented by King and Browning (1987) to quantify partisan symmetry in redistricting plans by assessing whether an electoral system treats political parties equally when their vote shares are hypothetically reversed. In essence, the metric evaluates asymmetries in the translation of acquired votes into congressional seats. By equal, it means that the reversed vote shares result in the same number of secured seats for each party. If the calculated partisan bias value equals zero, the electoral system satisfies the partisan symmetry standard, indicating that ideal partisan fairness is achieved and suggesting that partisan gerrymandering has not occurred. If the value is positive, then it indicates that the plan is favorable to the Democratic Party and vice versa. As given in Equation 3, partisan bias (β) is calculated by taking the statewide vote share of a political party ($S(V)$) and subtracting from it

the seat share that the party would have if the vote shares were reversed ($1 - S(1 - V)$). The value is then divided in half to represent the proportion of seats that would need to be reassigned between parties to restore symmetry (Katz et al., 2019).

$$(3) \quad \beta(V) = \frac{S(V) - [1 - S(1 - V)]}{2}$$

The judicial application of the partisan bias metric lies in the *League of United Latin American Citizens v. Perry* (2006), where SCOTUS considered it as quantitative evidence of gerrymandering but declined to adopt it as standard because there were no specific values which could be taken as threshold for what constitutes gerrymandering (Grofman & King, 2007). In *Gill v. Whitford* (2018), SCOTUS again evaluated arguments which used the partisan symmetry family of metrics, including partisan bias, but rejected their use as standalone evidence for determining whether gerrymandering has occurred (Ramchandran & Gold, 2018).

The partisan bias metric has several limitations. For example, partisan bias is inherently dependent on the construction of a seats–votes function, which requires assumptions about how votes would translate into seats under counterfactual vote swings. These assumptions are difficult to justify as empirically due to the electoral advantage of incumbent candidates and other factors such as the actual spatial distribution of voters and the existence of ethnic and language minority voter groups with preference for candidates from specific political parties (Katz et al, 2019). Furthermore, partisan bias belongs to the partisan-symmetry family of metrics which were all found to exhibit paradoxical outcomes when applied to actual electoral scenarios. Specifically, metrics such as the partisan bias may indicate that an election is perfectly fair and unbiased even when the vote-share outcomes exhibit strongly partisan outcomes (DeFord et al., 2023).

In terms of relevance for quantifying ethnolinguistic gerrymandering, partisan bias can only serve as a parallel indicator. As a metric of partisan symmetry, partisan bias does not give insight into the packing or cracking of ethnic and language minority voter groups on its own (Barton, 2022; Swamy et al., 2021). Furthermore, when used in isolation, metrics such as the partisan bias may lead to paradoxical outcomes where even clear examples of gerrymandering result in values of the metric which indicate that no gerrymandering has occurred (Ramachandra & Gold, 2018). Still, partisan bias carries benefits as a parallel indicator because it indicates for which political party a redistricting plan is advantageous (Swamy et al, 2021).

3.5.4 GEO Metric

Designed by Campisi et al. (2022), the Geography and Election Outcome (GEO) metric quantifies partisan gerrymandering by using both the geometry and electoral outcomes in districts. For the former, it models district adjacency to ensure that the vote share can only be swapped between adjacent districts. For the latter, GEO essentially calculates the number of lost districts that a party could flip to a 50/50 contest, called a competitive district, without losing any of the neighboring districts which it won. It was developed in response to *Rucho v. Common Cause* (2019) to utilize both geographic and election result data with the purpose of providing a deterministic and easily interpretable algorithm for use in legal cases.

As described by Campisi et al. (2022), the limitations of GEO lie in its dependance on vote shares rather than raw vote counts under the assumption that turnout is uniform across districts and their various voter blocks. In practice, this may not be true in areas of socio-economic, ethnic or language diversity as specific blocks may have higher turnout. Furthermore, GEO does not assess the validity of the alternative vote share swaps used to calculate the number of lost competitive districts for a party. As such, some of the calculated competitive districts may have implausible district shapes or any other characteristics which mean they would not be plausibly contested in

practice. Additionally, GEO requires accurate vote share and turnout data for each district and the geometry of the proposed district plans. Finally, the method does not consider differences in voting behavior between voter blocks, nor does it model ethnolinguistic bias (Campisi et al., 2022).

The strengths of GEO are the method's use of both geometry and voting outcome data which addresses a significant limitation of both compactness-based and other simulation-based methods which utilize only one of these types of data in their assessment of gerrymandering (Campisi et al., 2022). Despite the fact that GEO does not model ethnolinguistic bias, it may still contribute towards the development of an assessment method for ethnolinguistic gerrymandering as its core logic of calculating the number of lost competitive districts offers an alternative metric which is not found in other methods.

3.5.5 Areal Unit Segregation Measure

Conceptualized by Saporito and Maliniak (2022), the areal unit segregation measure (AUS) is an assessment method of ethnolinguistic gerrymandering to distinguish between redistricting plans which are naturally and unnaturally packed with racial minorities. It seeks to address the evidentiary expectations created by the *Bethune-Hill v. Virginia Board of Electors* (2018) court cases to evaluate whether a districting plan segregates racial minorities with the intent of affecting their electoral representation unconstitutionally through gerrymandering or follows natural demographic clustering through a simple mathematical model. The method is built specifically on addressing the MAUP, using measures of spatial segregation explored in Jakubs (1981) and Saporito (2017), by constructing population-weighted local environment of a fixed travel distance each resident within an actual redistricting plan. The district's total racial composition is then compared to the average of constructed local environments. If the AUS is 0, no racial gerrymandering has occurred while positive and negative values indicate packing and cracking respectively (Saporito and Maliniak, 2022). The equation is:

$$(4) \quad AUS = \pi_m - \frac{\sum t_p \pi_{pm}}{T}$$

Where:

- AUS: The segregation score of a legislative district
- π_m : The proportion of the district's total population that is part of group mmm (e.g., Black residents)
- t_p : The population at point p which is the centroid of a census block
- π_{pm} : The average proportion of group m in the local environment around point p
- T: The total population of the district DDD
- $\sum t_p \pi_{pm}$: A population-weighted average of the racial composition of the local environments surrounding all points in the district

The calculation of AUS is explicitly compatible with simulation-based assessment methods in the form of a metric to constrain on generated districts and to compare them with redistricting plans to identify outliers. The measure explicitly addresses the MAUP by constructing local environments of a fixed travel distance which may be further adjusted in other approaches (Saporito and Maliniak, 2022). Finally, being a metric is compatible with both algorithmic optimization approaches such as the SMC sampling used in DeFord et al. (2022) and simulation-based methods such as Powell et al. (2020).

The limitations of AUS lie in its data requirements which are high-resolution demographic data that require detailed census block geometries with data on racial composition rather than the more readily available precinct-level data. Additionally, the authors do not propose a fixed

threshold of what is constitutional packing, but instead simply identifies outliers and leaves their interpretation to legal professionals (Saporito and Maliniak, 2022). As such, the methods application to the redistricting plans scrutinized in the *Bethune-Hill v. Virginia Board Electors* (2018) follow standards established by the plaintiffs.

The utility of the AUS metric for the development of a new assessment method for ethnolinguistic gerrymandering lies both in its strengths and limitations. Being fundamentally a metric which can be simply computed given sufficient data, its ability to be integrated as a constraining factor in simulation-based approaches makes it potentially highly valuable. Additionally, the lack of a threshold for what is constitution packing established by the authors, considered a limitation of the method, can actually be its strength as it supports the approach of this research to not propose optimal redistricting plans, but to simply quantify gerrymandering through the lens of the MAUP.

3.6 Outlier Analysis: Implementation

Following the standard approach of outlier analysis, I generate an ensemble of potential, valid redistricting plans for the states of South Carolina and Texas. To facilitate this, I use the *GerryChain* package which is a Markov Chain Monte Carlo (MCMC) method that addresses the under- and oversampling limitation in the base approach to outlier analysis (MGGG Redistricting Lab, 2024). It addresses the limitation by creating a random walk through the map space where it picks samples to continue simulating the ensemble of redistricting plans (Ramachandran & Gold, 2018). To build potential districting plans, I use the Recombination Markov Chain (ReCom) algorithm which is a type of MCMC algorithm included in the *GerryChain* package. ReCom functions iteratively by merging adjacent districts within a state into a spanning tree and randomly repartitioning them while enforcing the following constraints for each generated district (DeFord et al., 2021):

1. Population equality: each district's total population must be within a set deviation relative to neighboring districts.
2. Contiguity: each generated district must remain a single connected object which contains adjacent precincts.
3. Compactness: avoids irregular district shapes and is facilitated by the use of spanning-tree cuts in the ReCom algorithm.

ReCom follows the first four steps of the general approach of outlier analysis as described in subchapter 3.3, but it deviates in the second half by following these steps: (5) Select two adjacent districts. (6) Merge the selected districts. (7) grow a random spanning three on the merged districts. (8) Cut one edge of the tree to create two contiguous districts of equal population. (9) Replace the original pair of districts from step 5 with the new pair. (10) Repeat until the specified number of unique plans is achieved.

Other than *GerryChain*'s code being available under open license, I select this package because it also facilitates the integration of new metrics in the form of updaters, which are functions applied to every finished redistricting plan in the ensemble (MGGG Redistricting Lab, 2024). As an output, *GerryChain* builds a data frame of values for the chosen metrics for the entirety of the ensemble without saving the boundaries drawn for every single plan in the ensemble unless specified in the parameters of the run. This is desirable as it reduces the necessary storage space and the required processing time. By default, the following metrics are included in *GerryChain* as updaters: efficiency gap, mean-median difference, and partisan bias. I used these three metrics in addition to adding the GEO Metric and AUS as updaters myself. In the case of the GEO Metric, I used the code provided in Campisi et al. (2022). There is no code available for the AUS, but the description of its calculation in Saporito and Maliniak (2022) is sufficient to recreate it as an updater.

Using *GerryChain*, I simulated ensembles of 10,000 unique plans each for the South Carolina's 2022 House Election and Texas 2026 election. I chose this value for the first parameter of unique

plans based on the general approach described for MCMC algorithms in Bangia et al. (2017), although no recommendations are given specifically for GerryChain (MGGG Redistricting Lab, 2024). For the second parameter, I used a fixed random seed to ensure consistent results on repeat run because the ReCom algorithm is stochastic. It is possible to run the algorithm without a fixed random seed, but it would lead to different results between runs and negatively impact reproducibility. For the third parameter, I choose a 1% tolerance for the population equality constraint following the recommendation of Ramachandran & Gold (2018). However, when the population deviation is calculated for PLANC2333 of Texas, it exceeds that value at 2.2%, which cannot be avoided. For the fourth parameter, I set the number of districts to 7 for South Carolina and 38 for Texas. For the final parameter, I specify that every thousandth plan in the ensemble must be saved for manual evaluation and comparison with the enacted redistricting plan.

To interpret the calculated values of the metrics, I follow the approach of Saporito and Maliniak (2022) and use the interquartile range of the ensemble to identify the outliers in the enacted redistricting plans of South Carolina and Texas. I flag any districts below or above 1.5 times the interquartile range as outliers.

4. Results

4.1 Systematic Literature Review

4.1.1 Challenges of Quantifying Gerrymandering

4.1.1.1 Inferring Discriminatory Intent in Gerrymandering

Several methods and metrics exist to quantify gerrymandering; however, not all meet the requirements set in *Alexander v. South Carolina State Conference of the NAACP* (2024), which determined that evidence of discriminatory intent must be given by the plaintiffs. Instead, methods such as compactness tests or metrics like the efficiency gap measure effect rather than attempting to infer intent (Stephanopoulos & McGhee, 2014; Swamy et al. 2025). Many existing methods and metrics used to quantify gerrymandering are suitable for identifying disparities in electoral results, but do not generate group-specific, quantitative evidence that ties enacted redistricting plans to discriminatory intent and ethnolinguistic bias in the redistricting process (Pardo, 2025). Consequently, these methods and metrics fall short of the evidentiary requirements set in *Alexander v. South Carolina State Conference of the NAACP* (2024).

Two types of general methods exist which may fulfill the requirements set in *Alexander v. South Carolina State Conference of the NAACP* (2024; Table 3): First, optimization methods seek to design an ideal redistricting plan according to weighted criteria chosen by the assessor. Mixed-integer linear programming models, such as the one described in Swamy et al. (2023), are capable of optimizing a redistricting plan for specific metrics, but the results of these methods have been framed as unable to meet the evidentiary requirements of the courts in *Rucho v. Common Cause* (2019) under the argument that these methods only demonstrate the preference of the modeler instead of identifying discriminatory intent. However, Dobbs et al. (2024) have shown that state geography and legal constraints are the leading factors which determine the outcome of optimal plan generation instead of modeler preference. Still, generating optimal maps under state-specific constraints does not address the evidentiary requirements set in *Alexander v. South Carolina State Conference of the NAACP* (2024) that are needed to infer discriminatory intent (Pardo, 2025).

Second, the heuristic methods generate great quantities of plans, which satisfy the criteria chosen by the assessor (Haas et al., 2020). This type of method is relevant to this research, as its strength in generating many feasible, comparable plans can facilitate the inference of intent through the comparison of simulated and enacted redistricting plans. An example of a heuristic method would be the Seed-Fill-Shift-Repair, which generates thousands of possible plans according to a state's redistricting criteria, which can be refined according to more complex demands of a state's legislature into plans ready for enactment (Haas et al., 2020). For examples of in-court application, the results of heuristic methods have been effectively given as evidence in court cases such as *League of Women Voters v. Commonwealth* (2018) to provide quantitative evidence of enacted plans constituting a gerrymander (Zhu, 2024). Rather than solely measure effect or suggest optimal redistricting, heuristic methods are strong candidates in addressing the requirements set by *Alexander v. South Carolina State Conference of the NAACP* (2024) because they create thousands of plausible plans following several generation constraints that enable later comparison or refinement.

Table 3. Categories of gerrymandering methods identified by Haas et al. (2020).

Characteristic	Optimization Methods	Heuristic Methods
Description	Uses objective functions with weighted criteria to identify a single “best” redistricting plan.	Constraint-satisfaction approach that builds multiple plans using predefined metrics without ranking them.
Judicial connection	Suggests an ideal plan which courts or commissions should adopt based on quantitative “optimality.”	Aligns with legal reasoning that accepts a range of valid plans; supports multiple perspectives on redistricting
Objective	Maximizes or minimizes a mathematical score derived from a redistricting evaluation criterion.	Generates a set of adequately different redistricting plans, which are legally and politically viable to support comparison.
Output	One optimal redistricting plan.	Many feasible plans that satisfy legal and technical constraints.
Strengths	Precise identification of ideal redistricting plan possible in theory.	Supports diverse, suboptimal redistricting plans and enables a comparison between them.
Limitations	Depends on chosen criteria, which may be arbitrary. No commonly accepted standard exists on which criteria are desirable in redistricting plans.	Reliant on the determination of constraints and human interpretation of what constitutes a standard redistricting plan versus an outlier.

Outlier analysis, which is the identification of data points that deviate significantly from other points, exemplifies the heuristic method as it involves the generation of thousands of plausible, valid redistricting plans that can be compared with enacted plans to identify whether they are outliers (Chen & Rodden, 2015; Aggarwal, 2017; Haas et al., 2020). The comparison is done using a set of metrics that evaluate partisan fairness, packing and cracking of voters, or geographic traits of the district boundaries in a plan. The logic of the method is that if redistricting plans of state legislatures are outliers compared to simulated plans according to the selected metrics, discriminatory intent can be inferred and used as evidence in gerrymandering court cases (Ramachandran & Gold, 2018). Outlier analysis is a suitable method for addressing the evidentiary requirements of the courts, as its outcomes were previously used in the Pennsylvania Supreme Court as the “most compelling evidence” to identify a redistricting plan as an unconstitutional gerrymander (*League of Women Voters v. Commonwealth*, 2018, p. 125; Zhu, 2024).

However, outlier analysis and its application to inferring discriminatory intent carry several limitations: First, the results are strongly dependent on the constraints chosen for plan generation, such as population equality threshold, compactness, country split rules and so forth, and whether these constraints can be justified academically and in the courts (Chen & Rodden, 2015, Pardo, 2025). Second, the identification of outliers depends on the chosen metrics and is impeded by the lack of specific threshold values, which can be used for this purpose. The lack of clear thresholds carries the risk that the identified cannot be justified under scrutiny, and when partisan fairness metrics such as the efficiency gap are chosen, the method should not be used to infer discriminatory intent and ethnolinguistic bias (Ramachandran & Gold, 2018). Third, the representativeness of the simulated ensemble as a neutral benchmark is questionable because districts are large areas with an large amount of possible ways in which precincts and census blocks can be split and aggregated, which means that outlier analysis must explicitly account for the modifiable areal unit problem if it is used to infer discriminatory intent in gerrymandering (Openshaw, 1983; Duchin & Walch, 2022; Barton & Eguia, 2024).

Outlier analysis is a suitable method for addressing the requirements set in *Alexander v. South Carolina State Conference of the NAACP* (2024) because it can generate group-specific, quantitative insights which tie enacted redistricting plans to discriminatory intent and

ethnolinguistic bias in the redistricting process (Ramachandran & Gold, 2018; DeFord et al., 2023; Pardo, 2025). By generating an ensemble of redistricting plans, which serves as a neutral benchmark because it was not created with ethnolinguistic bias, it is possible to compare enacted plans with the ensemble to identify whether they are outliers (Chen & Rodden, 2015). If an enacted plan is identified as an outlier, then it is possible to use the results of outlier analysis as supporting evidence of discriminatory intent and thus gerrymandering in the courts following the evidentiary bar set post *Alexander v. South Carolina State Conference of the NAACP* (2024).

4.1.1.2 Distinguishing Partisan and Ethnolinguistic Bias

Disentangling partisan and ethnolinguistic bias is the second requirement of *Alexander v. South Carolina State Conference of the NAACP* (2024), which most metrics cannot meet (Pardo, 2025). This limitation is exemplified in the family of partisan symmetry scores, which are gerrymandering metrics including the efficiency gap, mean-median difference, and partisan bias (Eguia, 2022). These metrics are commonly used in courts but are unsuitable for quantifying ethnolinguistic gerrymandering because they target only partisan bias without a spatial component (DeFord et al., 2023). They operate on district-level vote aggregates and seat–vote relationships and do not consider voting returns at a lower aggregation level nor do they consider ethnicity or language of the voters (Barton & Eguia, 2024). Most importantly, they are incapable of disentangling partisan and ethnolinguistic bias because they do not acknowledge the latter. Barton (2022) find that the efficiency gap, mean-median difference, and partisan bias can lead to values indicative of gerrymandering, even when ethnic and language minority voters are not packed or cracked in the district.

The efficiency gap is a frequently cited partisan symmetry metric due to its inclusion in *Gill v. Whitford* (2018) which sets precedent for the metric’s use in gerrymandering court cases (Ramachandran & Gold, 2018). It expresses the number of wasted votes by party which are in excess of what was needed to win a district (Stephanopoulos & McGhee, 2014). This metric carries two limitations that make it unsuitable for disentangling partisan and ethnolinguistic bias in gerrymandering: First, it is a race-blind metric that measures effect and therefore has no inputs related to voter ethnicity or language (Ramachandran & Gold, 2018; Zhu, 2024). This is by design as it is a metric free of ethnic classifications that is distinct from metrics which seek to measure compliance with the Voting Rights Act, which forbids the cracking and packing of voters based on their ethnic and language backgrounds (Stephanopoulos & McGhee, 2014; Ramachandran & Gold, 2018). In turn, one can argue that the efficiency gap does disentangle partisan and ethnolinguistic biases by focusing solely on the former. However, the interpretation of *Alexander v. South Carolina State Conference of the NAACP* (2024), supported by Pardo (2025), argues that using partisan-bias-only metrics does not constitute disentangling the two types of bias. As such, the efficiency gap metric does not meet the requirements set in *Alexander v. South Carolina State Conference of the NAACP* (2024).

The second limitation of the efficiency gap is its lack of a spatial component, as it uses only vote aggregates per party at the district level (Stephanopoulos & McGhee, 2014; Zhu, 2024). This is unsuitable for disentangling partisan and ethnolinguistic bias because voters of minority ethnic and language backgrounds tend to cluster in specific areas within a state, such as cities, where they can be packed or cracked within a compact district, which is fair by the standards of partisan metrics (Goedert et al., 2024). Using only district-level aggregates, the efficiency gap for two states would be the same even if in the most extreme cases of packing or cracking of ethnic and language minority voter groups (Chatterjee et al., 2020). This inability to distinguish between packing and cracking of minority voter groups by geographical clustering makes the efficiency gap ill-suited for disentangling partisan and ethnolinguistic biases or even to identify cases where ethnolinguistic gerrymandering occurs (DeFord et al., 2023). Moreover, the use of district-level

aggregates only makes the efficiency gap and other similar metrics highly susceptible to the MAUP.

AUS is suitable for quantifying ethnolinguistic bias because it analyses the clustering of ethnic and language minority voters and identifies whether a district is packed or cracked. This metric compares a district's share of minority voters to the share of its neighboring districts and thus gives quantitative evidence of ethnolinguistic bias in redistricting cases (Saporito & Maliniak, 2022). While not yet used directly in court cases, when applied to the redistricting plan discussed in *Bethune-Hill* (2019), the areal unit segregation measure showed that "8 of 12 districts Plaintiffs in *Bethune-Hill* claimed were packed had unusually (positive) AUS values" which indicates that these districts were packed with ethnic and language minority voters (Saporito & Maliniak, 2022, p. 15). Furthermore, when applied to the remedial maps suggested as a fairer alternative to the gerrymandered maps and accepted by the courts, AUS showed that in "10 of 11 redrawn districts the SM [Special Master appointed by the Supreme Court to redraw district boundaries] drew for the District Court, segregation was less than segregation in the districts ruled unconstitutional" (Saporito & Maliniak, 2022, p. 8). This use case shows that the areal unit segregation measure may be used to produce supporting quantitative evidence of ethnolinguistic bias. Furthermore, this metric disentangles partisan from ethnolinguistic bias as it does not require voting outcomes as inputs. As such the areal unit segregation measure meets the evidentiary requirements set in *Alexander v. South Carolina State Conference of the NAACP* (2024).

4.1.1.3 Analyzing Gerrymandering Through the Lens of MAUP

Redistricting is a spatial aggregation process, and thus, the modifiable areal unit problem (MAUP) must be understood when trying to infer discriminatory intent in gerrymandering cases. MAUP should be accounted for because both scale and zoning choices affect the outcome in the enacted plans, and similarly, these choices determine the results of the simulation process (Openshaw, 1983; Lee et al., 2025). The necessity of accounting for MAUP comes from the evidence that the scale of the basic units used in the redistricting process, whether they are blocks, precincts or voting tabulation districts, unpredictably affects the voting outcomes, demographic composition, and the values of the metrics in the resulting electoral districts (Openshaw, 1983; Saporito & Maliniak, 2022). Similar unreliability of the results is also the case in simulation-based redistricting analysis, such as the outlier analysis method, where using coarse spatial units or assuming homogenous distribution of voters within the chosen units will introduce MAUP-related bias in the outcomes of the used metrics (Powell et al., 2020).

The unpredictable influence of MAUP on the results of metrics when applied to redistricting plans suggests that all such measures should account for it, especially geographic metrics which quantify the geometry and shape of electoral districts (Sun, 2021). However, while these metrics are labeled as geographic, they do not account for MAUP because they study the district shapes instead of the aggregation process which creates them (Plane et al., 2019). Not accounting for MAUP is a strong limitation of these metrics as small alterations to the scale of input data will affect their values when simulating redistricting plans, e.g., with an outlier analysis, which makes them unreliable for use as a neutral benchmark or for drawing any conclusions when compared to enacted redistricting plans (Ramachandran & Gold, 2018; Lee et al., 2025).

Two types of geographic metrics illustrate the limitation of a metric that does not account for the effects of MAUP. First, the compactness tests and split counts are easy to compute and court-familiar, but they evaluate the shape of the districts and the number of counties, which were split to create them (Sun, 2021; Bowen, 2014). Compactness is typically quantified using measures such as the Polsby–Popper ratio, which compares a district's area to the area of a circle with the same perimeter (Dube, 2022). Alternatively, the Reock score can be used, which measures how closely a district fills its minimum enclosing circle (Sun, 2021). Both metrics describe only the shape of the district rather than the aggregation process and, therefore, may be unreliable indicators when

MAUP is a concern (Plane et al., 2019). Second, Buchanan et al. (2025) demonstrate that even less complex metrics, such as the county splits, are affected by MAUP. The county splits metric relies on the assumption that a redistricting plan is only valid when fewer counties are split than districts are created during the aggregation process. This assumption is unfounded, since Buchanan et al. (2025) showed that the split count is an arbitrary outcome of the aggregation choices and does not serve as an indicator of gerrymandering. The number of splits is determined by state geometry and population distribution, which means that zoning and scale choices, both elements of MAUP, are the determining factors and split counts are not indicative of gerrymandering. While the county split metric gives information regarding the aggregation process, it does not account for the MAUP and may be unreliable if used in a method such as outlier analysis.

The GEO metric improves on shape-only and county-split geographic metrics by explicitly combining geometry with district-level voting returns to count how many districts electoral outcomes can be flipped via boundary adjustments (Campisi et al., 2022). In essence, it recognizes that small changes to the aggregation process affect the resulting district boundaries and thus the electoral outcomes. However, the GEO metric does not use actual voter locations, which does not exist, and, hence, cannot account for the geographic clustering of voters. Therefore, the GEO metric cannot guarantee that implied swaps are always possible and will lead to the same outcomes in practice (Wise, 2025). Furthermore, the metric is gameable when used alone and should not be the only metric used to quantify gerrymandering (Ratliff et al., 2025). The GEO metric does not resolve MAUP in redistricting, but it is explicitly aware of MAUP and accounts for it in its calculation. Furthermore, GEO is suitable for integration in the outlier analysis approach because it gives insight into the degree of optimization for the outcomes of a specific party in an election and thus can be used to provide quantitative evidence of intent to gerrymander. However, the GEO metric does not disentangle partisan and ethnolinguistic bias on its own because it does not distinguish between the different ethnic and language background of the voter (Campisi et al., 2022).

4.1.2 Recommendation for Quantifying Ethnolinguistic Gerrymandering

To infer discriminatory intent while disentangling partisan and ethnolinguistic bias to meet the requirements of *Alexander v. South Carolina State Conference of the NAACP* (2024), outlier analysis can be paired with ethnolinguistic and MAUP-aware metrics. Specifically, the Areal Unit Segregation ($S^+/-$) measure detects packing and cracking of ethnic and language minority voters per district, and the GEO metric which to link those local patterns to district-level electoral consequences. Evaluated across a neutral ensemble, an enacted plan that is a consistent outlier and exhibits concentrated $S^+/-$ patterns aligned with GEO-implied seat effects provides group-specific, quantitative evidence which may support the inference of discriminatory intent and ethnolinguistic bias.

Table 4. Overview of gerrymandering assessment methods discussed in the preliminary literature review.

Method	Type of bias	Indicator	Data requirements	Strengths	Limitations
Polsby-Popper / Reock Compactness Scores	NA	District shape irregularity	District geometries	Simple to calculate and interpret; widely used in courts	Does not consider demographics or voter behavior; manipulable with modern computing
Geography and Election Outcome (GEO)	Partisan	Number of flipped competitive districts	Vote shares, district geometries	Uses both geographic and electoral outcome data	Assumes uniform turnout; ignores ethnolinguistic effects
Efficiency Gap	Partisan	Difference in wasted votes between parties	District-level vote totals	Clear threshold; legally recognized	Assumes 50% ideal share; no spatial or demographic components
Mean–Median Difference (MMD)	Partisan	Vote share skew	District-level vote shares	Simple and intuitive	Does not include spatial or demographic data
VRA-Constrained Ensemble Simulations	Partisan	Seat outcome deviation from simulated baseline	Precinct-level votes, population, race data, district geometries	Can distinguish intent; supports legal analysis	Computationally intensive; requires high-res data
Areal Unit Segregation Measure (AUS)	Ethnolinguistic	District vs. local environment racial composition	Census block demographics, district geometries	MAUP-aware; compatible with simulations; identifies packing/cracking	No threshold for gerrymandering; high data demands

4.2 Outlier Analysis: Results

4.2.1 South Carolina Midterm Elections 2022

Figure 7 shows the chain diagnostics of the simulated ensemble compared to the plan S.865 which was the focus of *Alexander v. South Carolina State Conference of the NAACP* (2024). For comparison, three implementations were selected from the chain based on the number of seats secured by the Democratic Party. Implementation 0 is the chain's random seed and the first implementation in which the Democratic Party attains no seats. Implementation 341 was selected because it is the first case in the chain where 2 seats are attained by the Democratic Party. Implementation 9999 has 1 seat for the Democratic Party and thus completes the overview of all possible scenarios based on the seat distribution. For the entire ensemble, the mean number of seats won by the Democratic Party equals 0.47, which indicates that the 0-seat outcome is the most common, as secured seats are discrete numbers.

First, the number of cut edges, which is unitless, averages 421.1 across the ensemble, while S.865 requires 474 cut edges, implementation 0 needs 550, and implementations 341 and 9999 require 419 and 392, respectively. Second, the ensemble's maximum population deviation per implementation averages 0.866%, whereas S.865 requires a deviation of 0.574%, which falls within 1.5 times the interquartile range. The selected implementations show deviations of 0.746% for run 0, 0.676% for 341, and 0.874% for 9999. Plan S.865 is not identified as an outlier in either diagnostic.

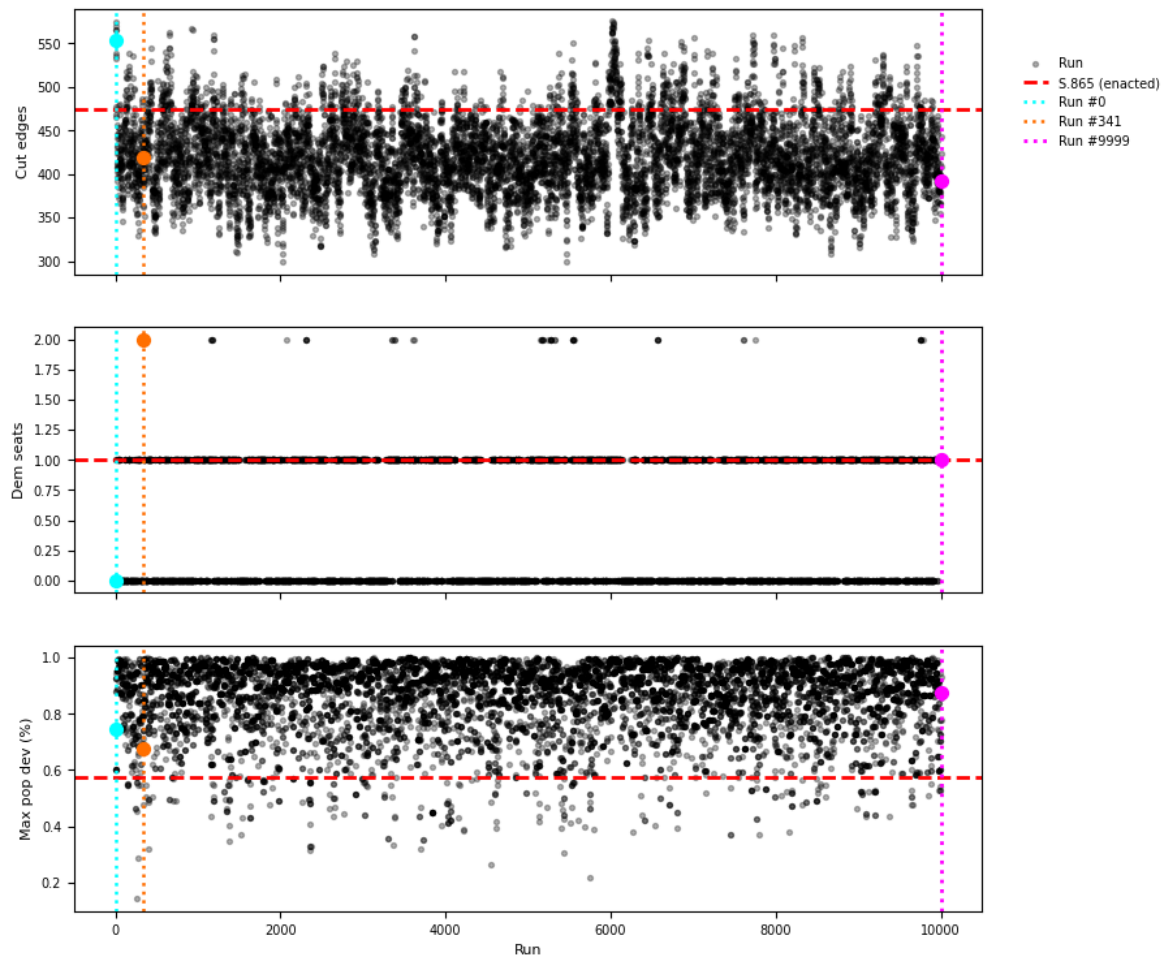


Figure 7. Diagnostics of the redistricting simulation for the South Carolina 2022 Midterm election. The enacted plan refers to S.865, while implementation refers to the individual steps of the Markov Chain within the simulated ensemble.

Figure 8 panel (a) shows several differences between the enacted plan S.865 and the three selected implementations from the simulated ensemble. For example, the edges of districts in Plan S.865 are less jagged, however, the districts appear less compact. Based on visual inspection alone, Figure 8 panel (a) does not show evidence that packing or cracking occurred. The westernmost districts in S.865 only appear to be Republican Party-dominated because no Democratic Party candidates ran in those areas.

In contrast, the selected ensemble implementations shown in Figure 8 panels (b-d) exhibit more irregular and jagged district boundaries, which is to be expected considering that they were generated through random recombination of voting tabulation districts. The westernmost districts no longer have a 100% vote share for the Republican Party because they now include areas with Democratic candidates.

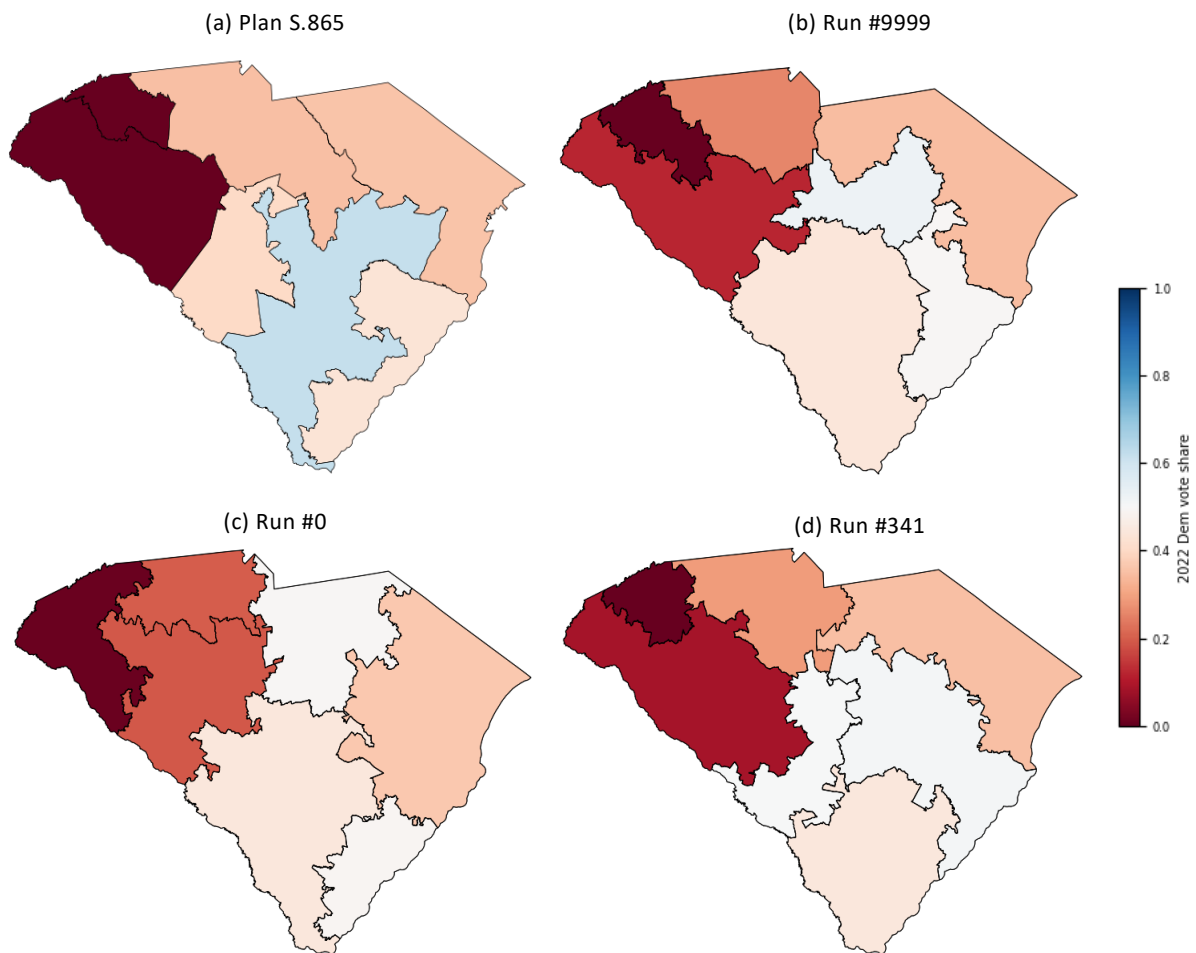


Figure 8. Electoral outcomes of the enacted plan S.865 and three simulations for the South Carolina 2022 Midterms: (a) S.865: Democratic seats = 1, Republican seats = 6, (b) Implementation 9999: Democratic seats = 1, Republican seats = 6, (c) Implementation 0: Democratic seats = 0, Republican seats = 7, (d) Implementation 341: Democratic seats = 2, Republican seats = 5.

Figure 9 shows the efficiency gap values for S.865 and the three selected ensemble implementations. Plan S.865 has a value of -0.022 , compared to -0.010 for implementation 9999, -0.158 for implementation 0, and 0.154 for implementation 341. The ensemble's distribution spans a wide range across positive and negative values, and the enacted plan lies well within this range. Based on this metric alone, S.865 cannot be identified as an outlier.

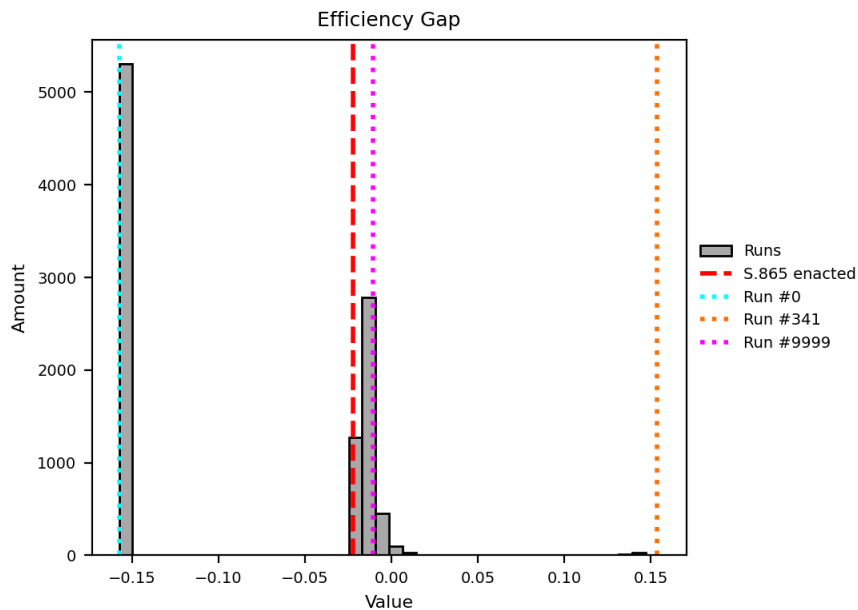


Figure 9. Distribution of efficiency gap values for the simulated ensemble and selected implementations.

Figure 10 shows the mean–median difference metric for S.865 and the three selected implementations. S.865 has a value of -0.022 , compared to -0.158 for implementation 0, 0.154 for implementation 341, -0.010 for implementation 9999, and an ensemble mean of 0.056 . The enacted plan's value lies within 1.5 times the interquartile range of the ensemble. Based on the efficiency gap metric, S.865 cannot be identified as an outlier.

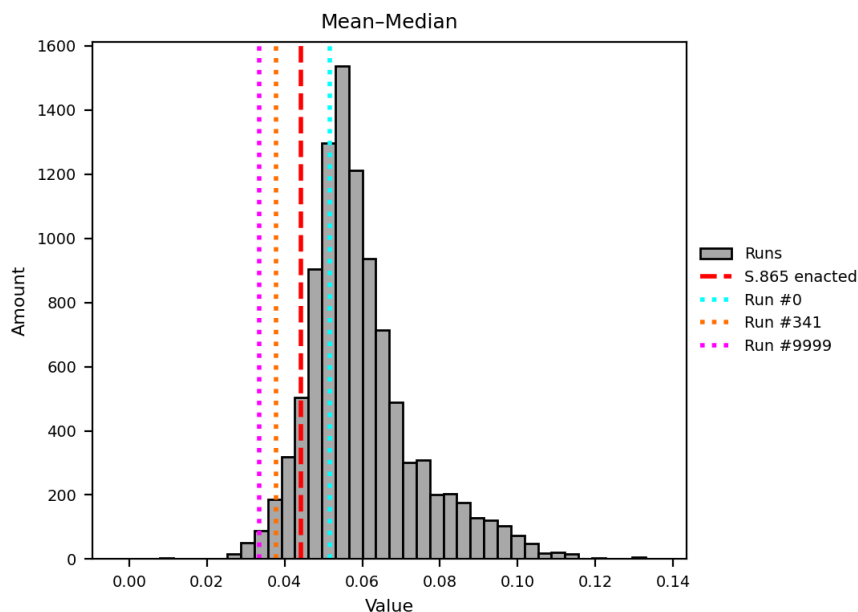


Figure 10. Distribution of the mean–median difference values for the simulated ensemble and selected implementations.

Figure 11 shows the partisan bias values for S.865 and the three selected ensemble implementations. The enacted plan has a value of 0.214, which is unitless, while implementations 0, 341, and 9999 each have the same value of 0.071. Although the value for S.865 is higher than the selected implementations, the ensemble distribution has a majority of its implementation also at the value of 0.214. Therefore, plan S.865 does not fall outside the range of values calculated for the simulated ensemble and it cannot be identified as an outlier using solely the partisan bias metric.

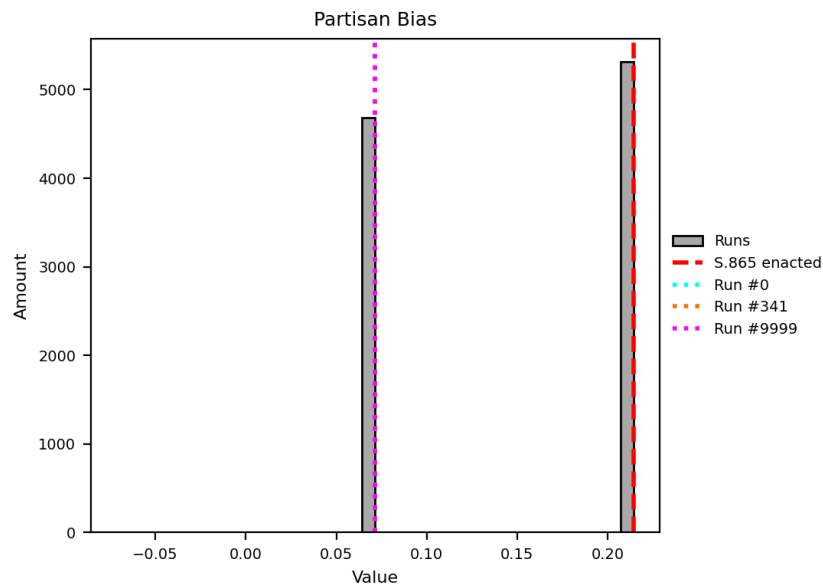


Figure 11. Distribution of the partisan bias values for the simulated ensemble and selected implementations.

Figure 12 shows the GEO metric values, again unitless, for S.865 and the selected ensemble implementations. Figure 12 Panel (a) assigns the Democratic Party a value of 2 for plan S. 865, compared with 1 for implementation 9999, 0 for implementation 341, and 3 for implementation 0. Since the values of the GEO metric are discrete, the interquartile range is not applicable, but the value of plan S.865 still falls well within the range produced by the ensemble and cannot be identified as an outlier using the GEO metric. The metric indicates that S.865 is more advantageous for the Republican Party, as it has a lower value of 1 than 2 for the Democratic Party.

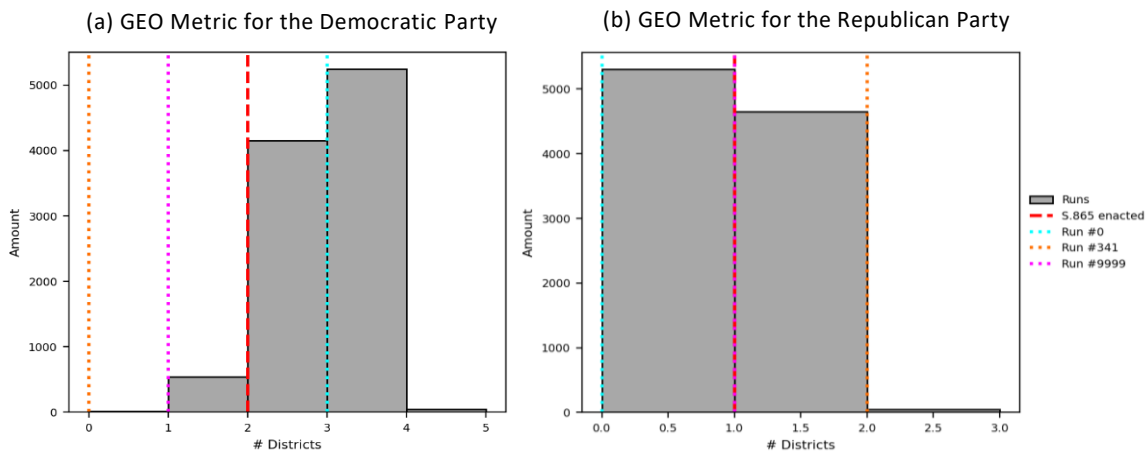


Figure 12. Distribution of the GEO metric values for the simulated ensemble and the selected implementations: (a) Values for Democratic Party, (b) Values for the Republican Party. Higher values indicate that a plan is more optimal for the given party.

Error! Reference source not found. shows the average areal unit segregation measure (AUS) values for S.865 and the selected ensemble implementations. Plan S.865 has an AUS value of -0.381 , compared with -0.752 for implementation 0 and -0.749 for implementations 341 and 9999. This makes the enacted plan a clear outlier when averaged across the entire state.

When analyzed by district, Figure 14 indicates that severe packing has occurred in district 6, which includes the Democrat Party-dominated area of the city of Columbia, SC. In Figure 8 panel (a), it is also the only district won by the Democratic Party, likely due to the packing of minority ethnicity and language voters. In comparison, all the surrounding districts appear to be slightly cracked. When compared to implementation 9999 in Figure 15, it appears that packing is more severe in S.865, however, cracking also occurs but to a lesser degree.

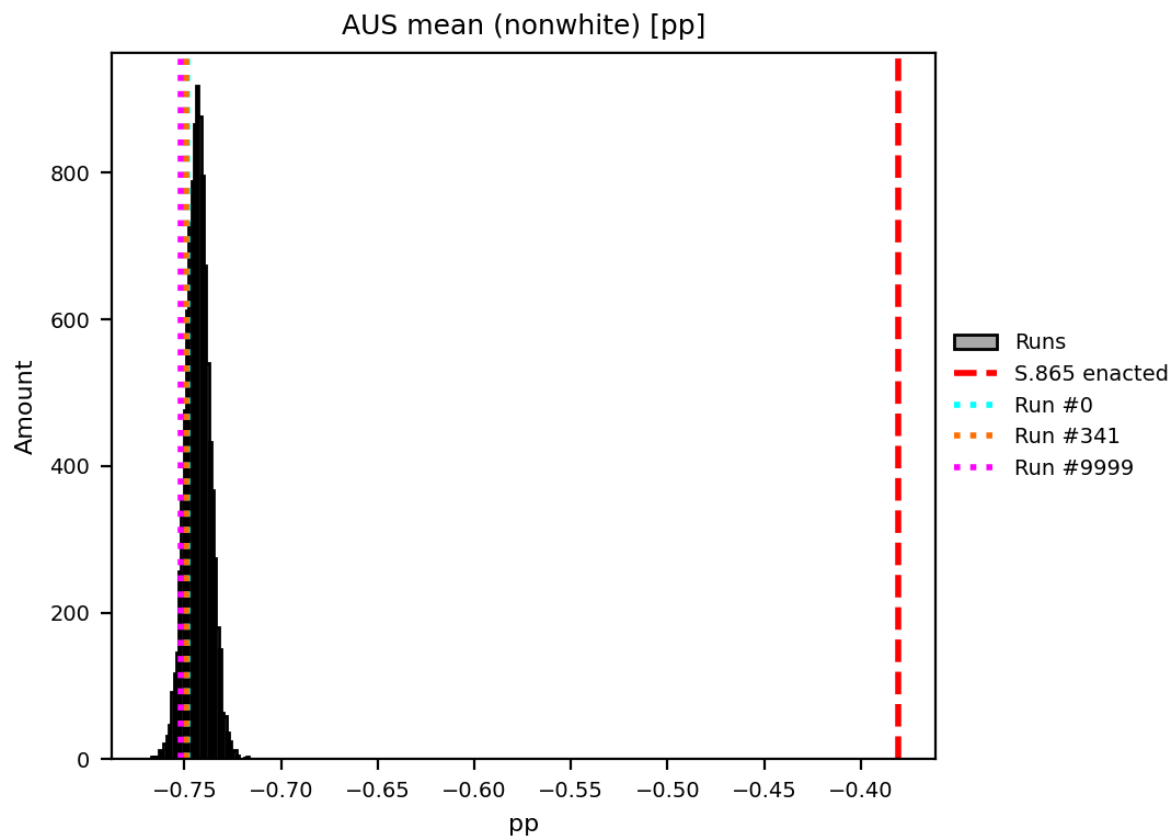


Figure 13. Distribution of the areal unit segregation value means calculated for the simulated ensemble and selected implementations.

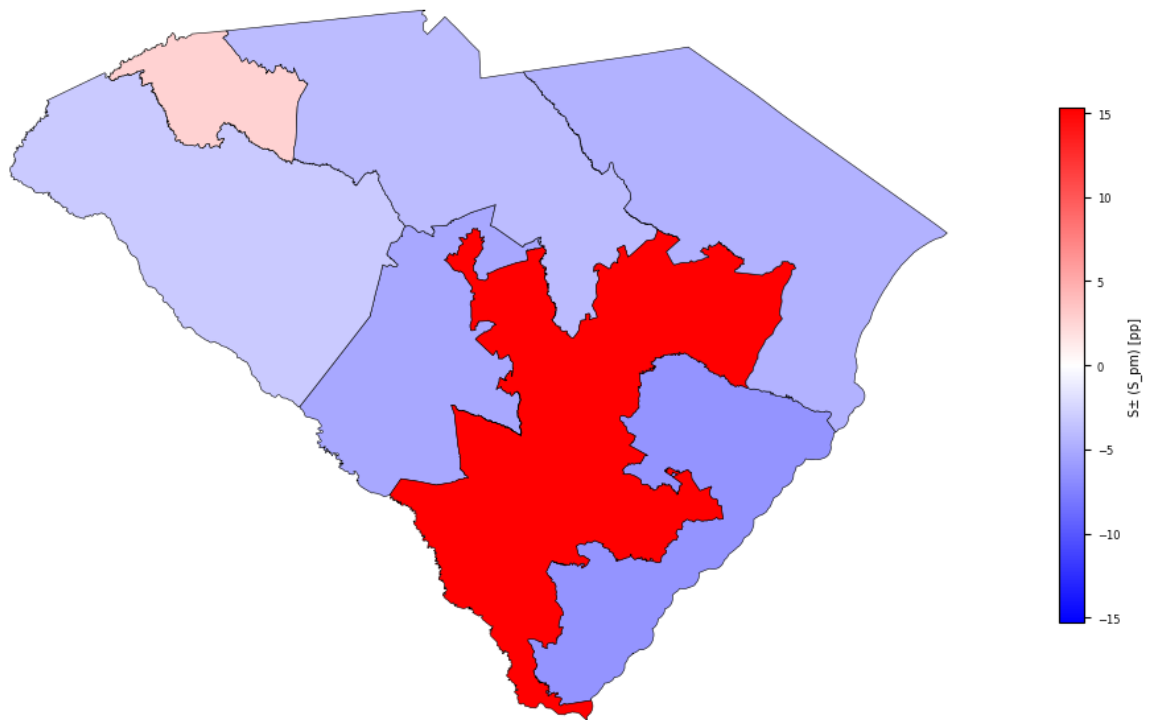


Figure 14. District-level AUS values for plan S.865. Negative values (blue) indicate cracking. Positive values (red) indicate packing.

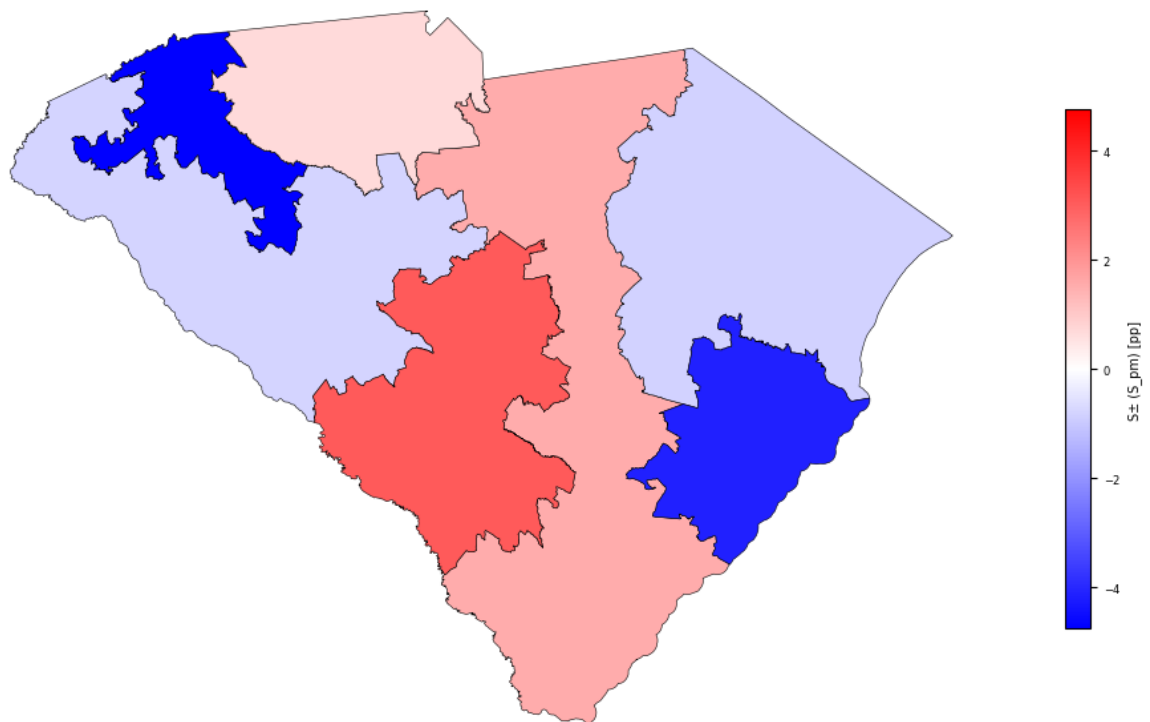


Figure 15. District-level AUS values for implementation 9999. Negative values (blue) indicate cracking. Positive values (red) indicate packing.

4.2.2 Texas Midterm Elections 2026

The Texas 2026 Midterm election provides a contrasting case to South Carolina, as its enacted redistricting plan, PLANC2333, is an outlier compared to the simulated ensemble. Figure 16 shows the diagnostics of the simulated ensemble compared to PLANC2333. First, the number of cut edges, which is the count of voting tabulation districts that are split when districts are formed, averages 2565.4 across the ensemble, whereas PLANC2333 requires 2979 cut edges, a 16.2% increase relative to the mean. Second, the mean number of seats won by the Democratic Party across the ensemble is 17.56, whereas for PLANC2333 it is 10. Third, the ensemble's maximum population deviation per implementation averages 0.97%, consistent with the imposed 1% population-equality threshold for the simulation, while PLANC2333 requires a deviation of 2.2%, which exceeds the threshold selected for the ensemble.

According to all three simulation diagnostics, the enacted plan is a significant outlier, falling outside the 1.5 times the interquartile range for the cut edges, seats won by the Democratic party, and the maximum population deviation. However, in the case of the maximum population deviation, this outlier does not suggest that gerrymandering has occurred because population deviations higher than 1% are also found in districts where gerrymandering has not occurred. The interpretation of the population deviation in the context of identifying gerrymandering is examined further in Chapter 5.

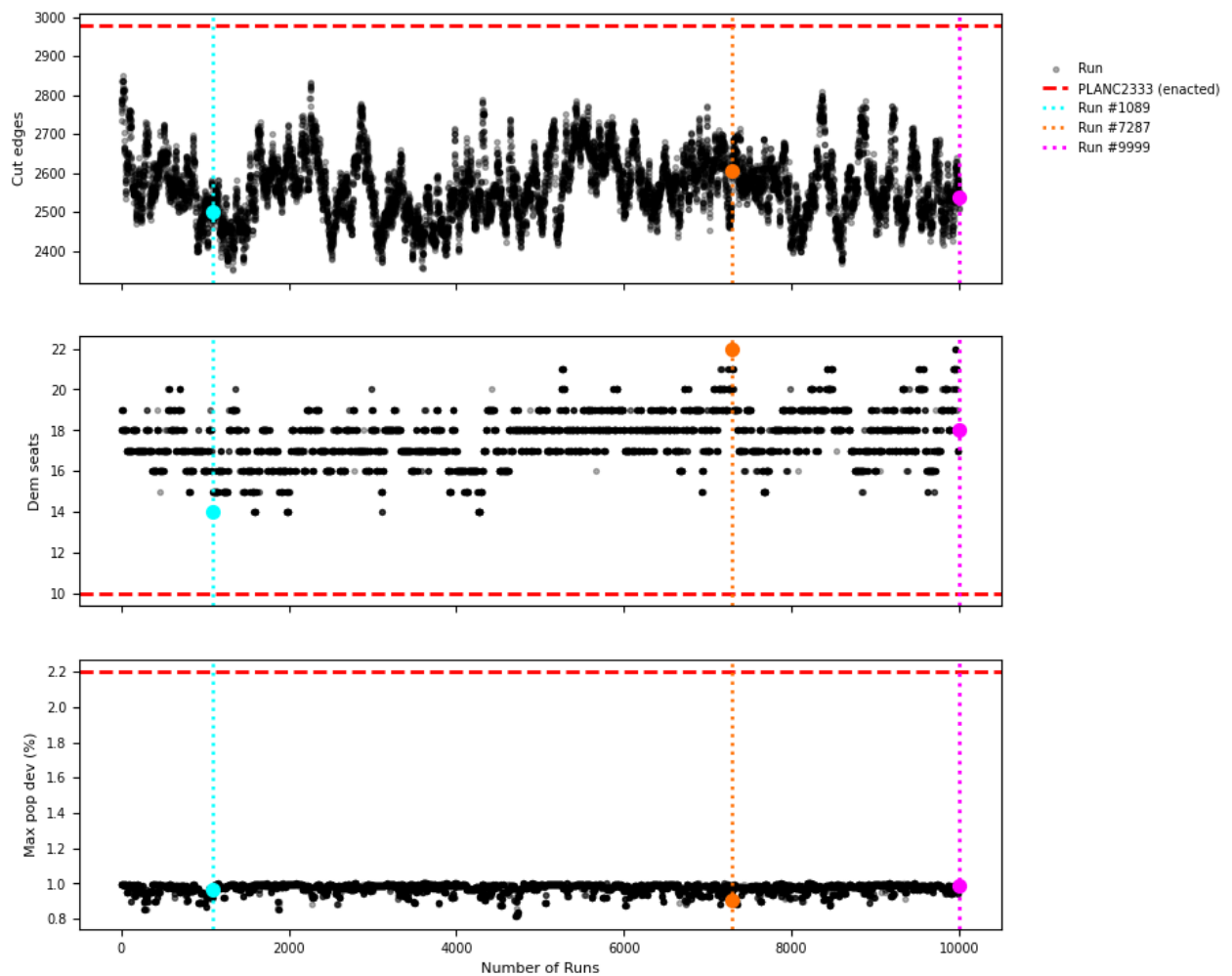


Figure 16. Diagnostics of the redistricting simulation for the Texas 2026 Midterm election. The enacted plan refers to PLANC2333 while implementation refers to the individual steps of the Markov Chain within the simulated ensemble.

PLANC2333 differs in several regards from implementation 1089, 7287, and 9999 of the simulated ensemble. A visual comparison of PLANC2333 and implementation 9999 in Figure 17 (a) shows that PLANC2333 repeatedly combines portions of the Democratic Party-dominated urban areas with large tracts of Republican Party-dominated rural territory. This urban-rural combination produces elongated districts in which the Republican Party wins by narrow but reliable margins, a pattern characteristic of cracking. At the same time, PLANC2333 creates small, highly compact districts that include only the urban areas of Austin, Houston, and the Dallas–Fort Worth metroplex, where the Democratic Party wins decisively, which is characteristic of packing. By applying both cracking and packing in PLANC2333, the Republican Party wins 28 seats while the Democratic Party only secured 10.

In contrast, the selected ensemble implementations displayed in Figure 17 (b-d) all show more compact, rotund districts with jagged edges despite differing strongly in the number of seats won by the Democrat Party with the minimum being 14 seats and maximum 22. Only PLANC2333 in Figure 17 (a) with 10 seats secured by Democratic Party displays less-compact districts where urban-rural combination results in victories for the Republican Party. This comparison highlights that only PLANC2333 departs from the geometric patterns found across the ensemble, both in shape and in its lower number of seats secured by the Democratic Party.

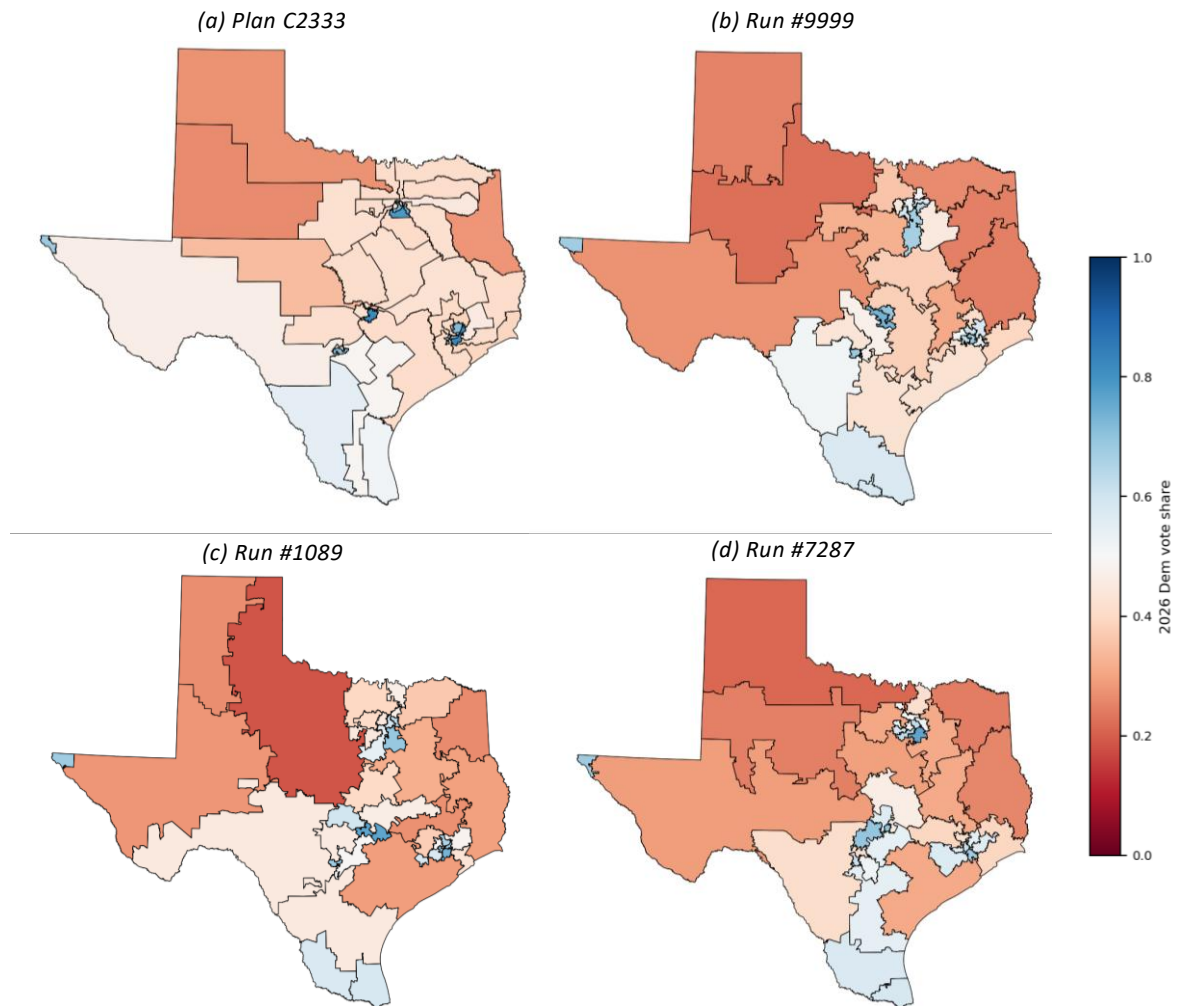


Figure 17. Electoral outcomes of the enacted plan PLANC2333 and three simulated implementations for the Texas 2026 Midterms: (a) PLANC2333: Democratic seats = 10, Republican seats = 28, (b) Implementation 9999: Democratic seats = 18, Republican seats = 20, (c) Implementation 1089: Democratic seats = 14, Republican seats = 23, (d) Implementation 7287: Democratic seats = 22, Republican seats = 15. For the legend, a value of 1.0 (dark blue) indicates that the Democratic Party secured 100% share of the vote.

Figure 18 shows the values of the efficiency gap metric, for PLANC2333 and the three selected implementations from the ensemble. PLANC2333 has a value of -0.220 , compared to -0.019 for implementation 1089, -0.003 for implementation 9999, 0.038 for implementation 7287, and an ensemble mean of 0.011 for the efficiency gap metric. The interquartile range for the ensemble spans -0.089 and 0.066 , and the efficiency gap calculated for the enacted plan falls outside of this range, making it an outlier. In contrast, all three selected implementations fall neatly within 1.5 times the interquartile range.

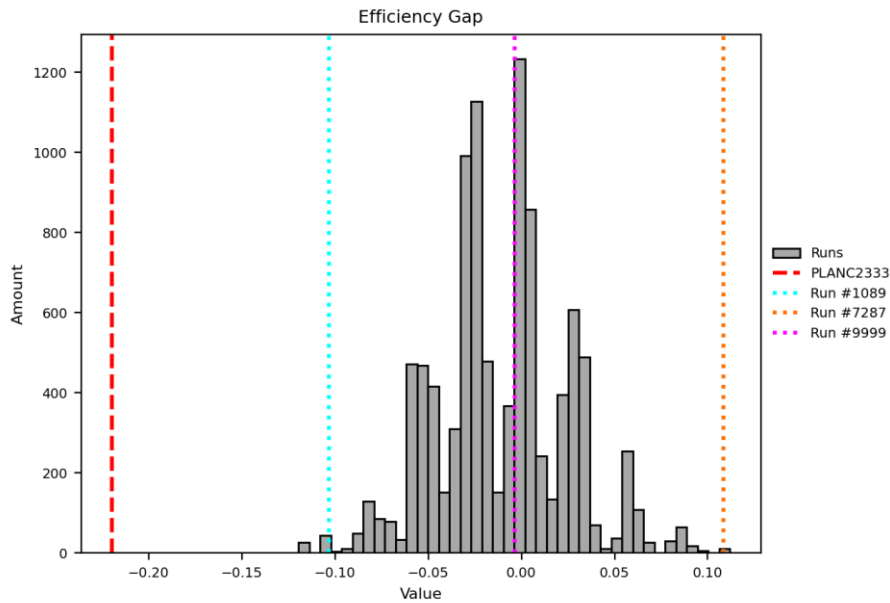


Figure 18. Distribution of efficiency gap values for the simulated ensemble and selected implementations.

Figure 19 shows the mean-median difference metric for PLANC2333 and the three implementations. PLANC2333 has a value of -0.063 , compared to -0.019 for implementation 1089, 0.007 for implementation 9999, 0.038 for implementation 7287, and an ensemble mean of 0.004 . The ensemble's interquartile range spans -0.034 to 0.042 , and the enacted plan's value falls outside this range, making PLANC2333 a significant outlier under this metric unlike the three selected implementations.

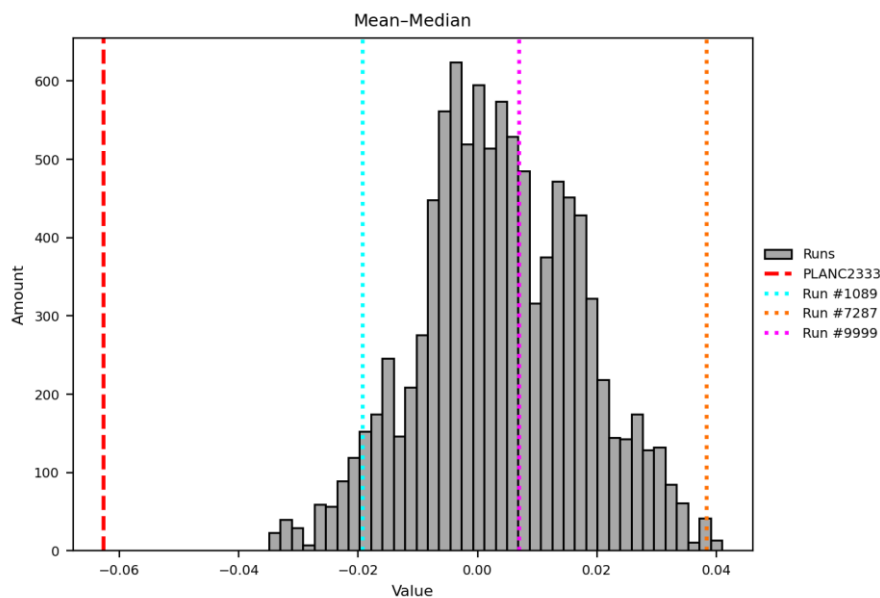


Figure 19. Distribution of mean-median difference values for the simulated ensemble and selected implementations.

Figure 20 gives the partisan bias values for PLANC2333 and the three selected implementations. The enacted plan has a value of -0.184 , compared to -0.053 for implementation 1089, 0.0 for implementation 9999, and 0.079 for implementation 7287, and an ensemble mean of 0.011 . The ensemble's interquartile range spans -0.039 to 0.066 , and the enacted plan's value lies outside this interval, making PLANC2333 a clear outlier under the distribution of the partisan bias metric.

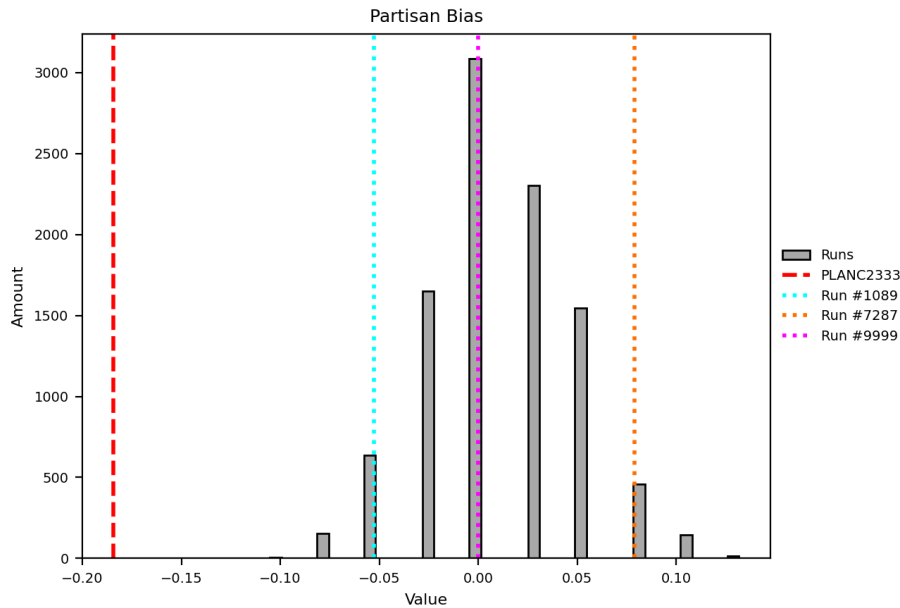


Figure 20. Distribution of partisan bias values for the simulated ensemble and selected implementations.

Figure 21 shows the GEO metric values for PLANC2333 and the selected ensemble implementations. In Figure 21 panel (a), the Democratic Party has a value of 16 for PLANC2333, which means that they could flip 16 additional districts through small vote-share transfers, compared to just 9 in implementation 9999 where they already won 18 seats. In implementations 1089 and 7287, the GEO metric is 6 and 14 respectively. Although the GEO values are discrete and therefore an interquartile range is not applicable, PLANC2333's values remain outside the range observed across the ensemble, identifying it as a clear outlier under this metric. In Figure 21 (b), the GEO metric for the Republican party is also extremely low compared to the values of the ensemble, which makes PLANC2333 an outlier, because the plan appears optimized in favor of the Republican Party.

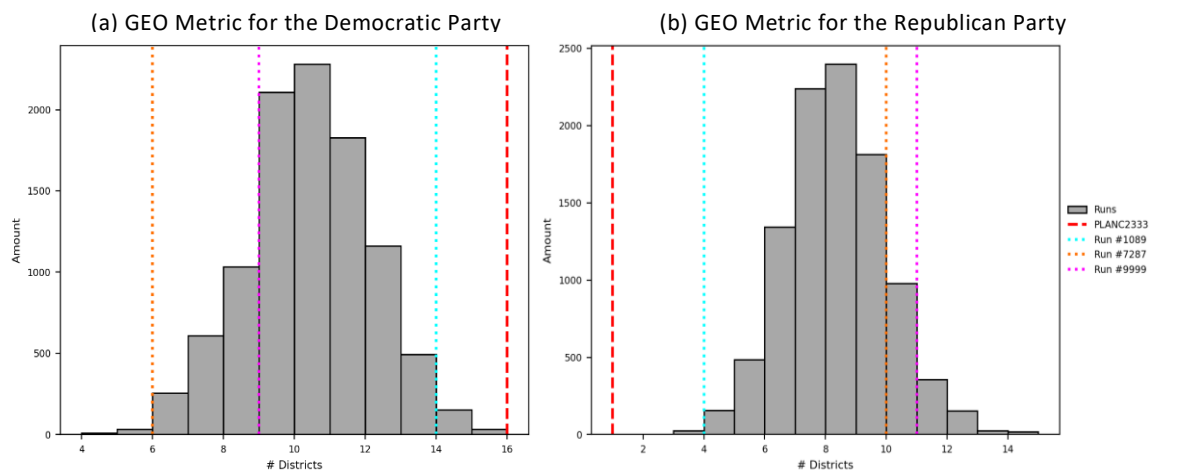


Figure 21. Distribution of the GEO metric values for the simulated ensemble and the selected implementations: (a) Values for the Democratic Party, (b) Values for the Republican Party. Higher values indicate that a plan is more optimal for the given party.

Figure 22 shows the average areal unit segregation measure (AUS) values for PLAN C2333 and the selected ensemble implementations. PLAN C2333 has an AUS value of -1.326 , which is equal to the value of implementations 1089 and 7287. Only implementation 9999 differs from the values of -1.322 ; the overall ensemble mean is -1.325 . All these values lie within the ensemble's interquartile range, which is -1.355 to -1.316 , i.e., PLAN C2333 is not an outlier when AUS is averaged across the entire state.

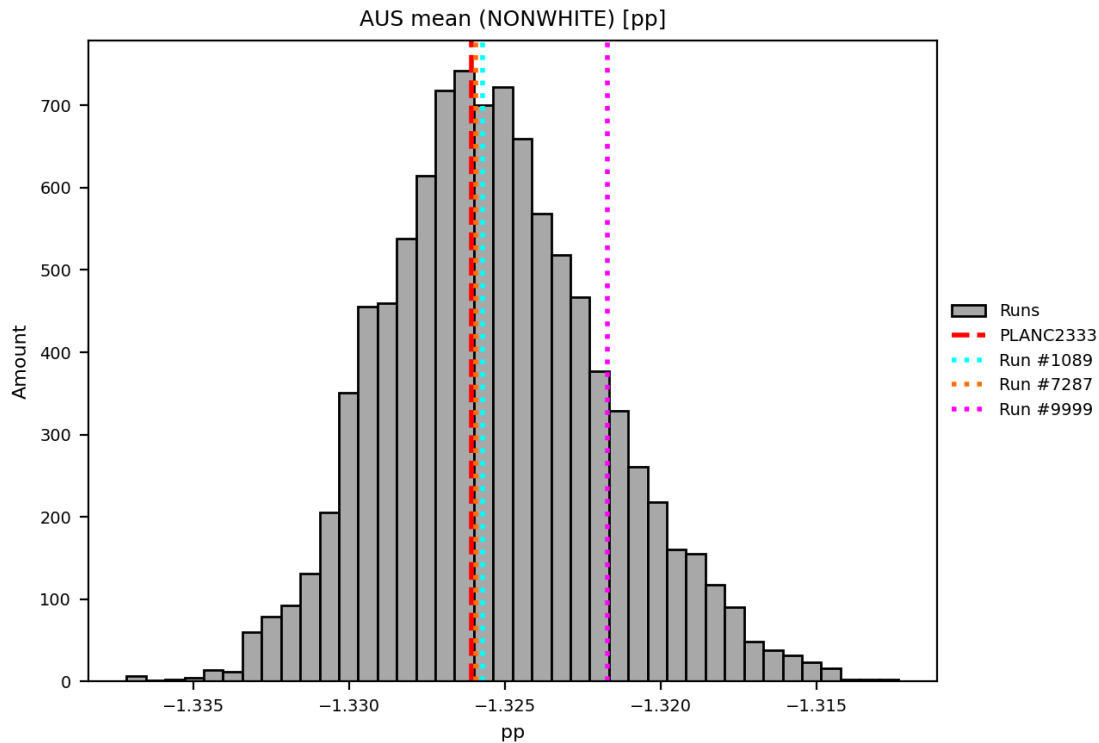


Figure 22. Distribution of the areal unit segregation value means calculated for the simulated ensemble and selected implementations.

However, as Figures 23 and 24 show, the average masks important local deviations around the major metropolitan areas. As visible in Figure 23, districts surrounding the Democratic-leaning urban centers of Austin, Houston, and the Dallas–Fort Worth metroplex display more negative AUS values (dark blue), indicating that ethnic and language minority voter districts, grouped under the label of non-white, are cracked across adjacent districts. In contrast, the urban cores themselves exhibit highly positive AUS values (dark red), suggesting that minority voters are packed into a small number of districts.

Compared to PLAN C2333, Figure 24 shows that the patterns of cracking and packing are not identifiable in implementation 9999. In implementation 9999, the districts directly adjacent to these metropolitan areas display AUS values closer to zero (lighter blue), which indicates less severe cracking of minority populations. No packed districts (dark red) can be identified.

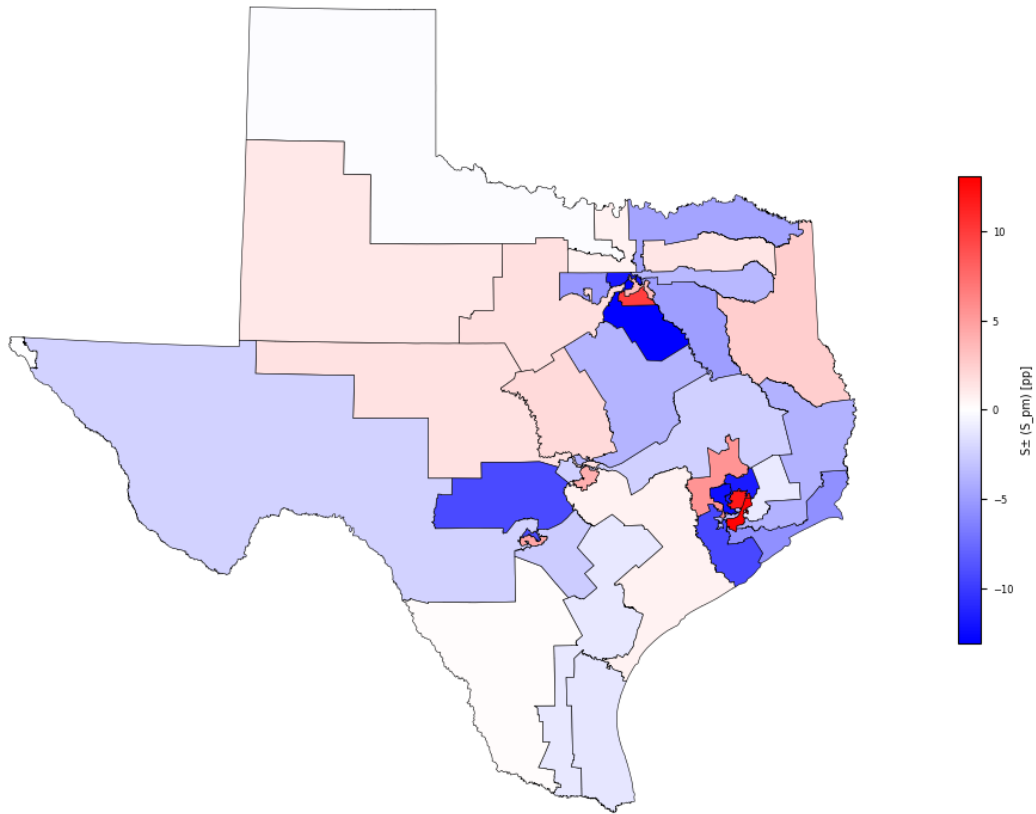


Figure 23. District-level AUS values for PLAN2333. Negative values (dark blue) indicate cracking. Positive values (dark red) indicate packing.

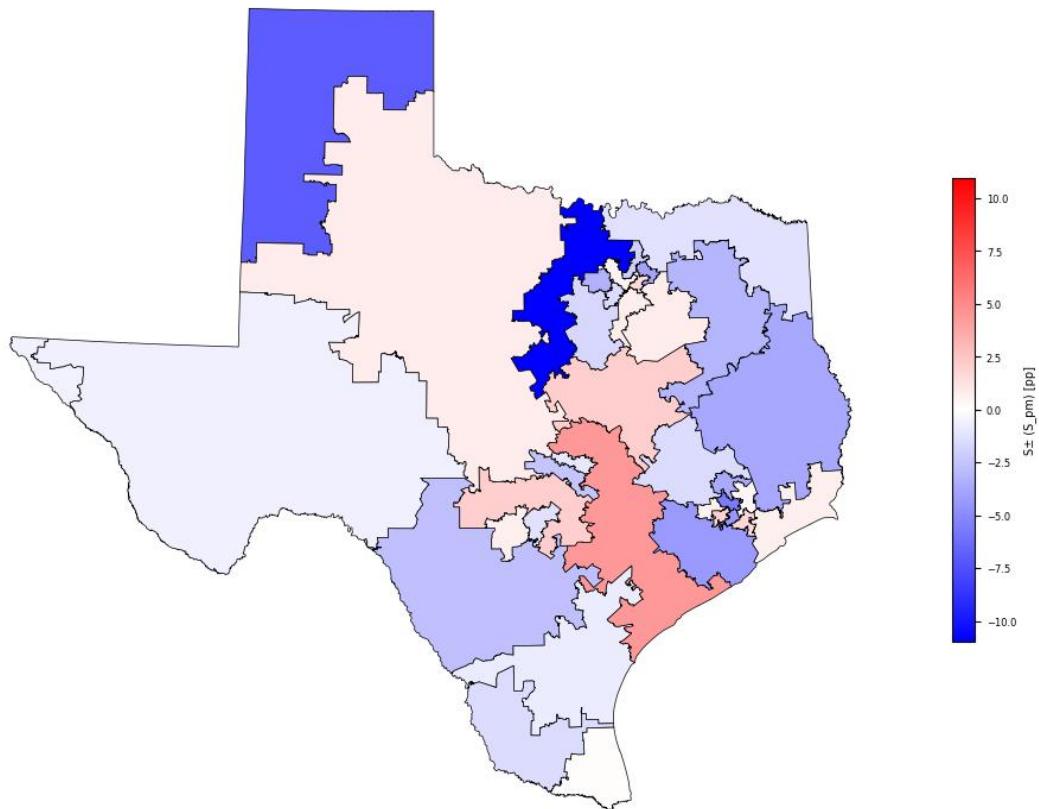


Figure 24. District-level AUS values for implementation 9999. Negative values (dark blue) indicate cracking. Positive values (dark red) indicate packing.

4.2.3 Sensitivity Analysis

I conducted a sensitivity analysis to assess whether the parameters chosen for the Texas and South Carolina simulations determine the findings described in the previous two subchapters. The results of the analysis suggest that the findings are robust and show no significant differences in the chain diagnostics and calculated metrics based on parameter choices, with the sole exception of the ideal number of districts. Across all simulations, I kept the same seed to ensure consistent results.

First, the total number of steps in the Markov Chain, which is the total number of implementations created for the ensemble, has no effect on the patterns visible in the chain diagnostics. Even when set to 40,000 steps, no patterns emerge in the figures for cut edges, seats won by the Democratic Party, nor the maximum population deviation. It does not shift the calculated ensemble means, interquartile ranges, nor the placement of the enacted plans within the distributions. This lack of new patterns confirms that 10,000 steps is an adequate number of implementations for the analysis of the Texas 2026 Midterms and South Carolina 2022 Midterms.

Second, raising the maximum population-equality tolerance from 1% to 2.2%, which is needed to draw the Texas PLANC2333 plan, does not change the distribution for the five calculated metrics. The mean seat outcomes, the cut-edge distributions remain consistent, however, the maximum population deviations calculated for the ensemble are more in-line with PLANC2333 as expected. The sole change is to the processing time of the ensemble generation which becomes shorter. The lack of changes to the distribution of the calculated metrics suggests that a maximum population-equality tolerance of 1% is an appropriate choice, which is also recommended by Ramachandran & Gold (2018).

Third, changing adjacency rules from rook contiguity to queen contiguity has no recognizable effect on the chain diagnostics and distribution of calculated metrics. While there appear to be no effects for the states of South Carolina and Texas, a different pattern could be found for states with unique shapes such as Florida, New York State, or Maryland where voting tabulation districts are less equally connected within the state's boundaries. However, this falls outside the scope of the study.

Fourth, the ideal number of districts proved very impactful. Raising the number of districts to be drawn for Texas and South Carolina drastically changed the chain diagnostics and the distribution of the calculated metrics, which is expected. Changing the number of districts in Texas from 38 to for example 7 affects all results. However, this invalidates any comparison with the enacted plans S.865 and PLANC2333 because both states have statutory district counts. Comparing a simulated 7-district with PLANC2333 which follows the strict 38-district expectations is simply not a valid approach to analyzing redistricting.

Still, an implication of the ideal district-count effect is that comparing states with drastically different numbers of districts may lead to unfounded conclusions. A 7-district state will naturally lead to a different distribution of values for gerrymandering metrics than a 38-district state. As such, gerrymandering assessment should not be conducted by comparing different states with each other. Instead, assessment should be carried out on a state-by-state basis or only between states with similar statutory district counts.

5. Discussion

5.1 Discussion of the Results

5.1.1 Comparison of Gerrymandering Metrics in the Context of MAUP

The application of the five gerrymandering metrics yielded contrasting results both across states and metrics for South Carolina's 2022 and Texas' 2026 midterm elections. Consistent with DeFord et al. (2023) and Ratliff et al. (2025), outcomes differ across metrics even when applied to the same election. When evaluated for their ability to provide quantitative insights into ethnolinguistic gerrymandering, the key discrepancy between metrics is in the areal unit segregation measure (AUS; Saporito & Maliniak, 2022) when compared to the other four. In contrast to DeFord et al. (2023), this study finds no differences between the partisan symmetry metrics (efficiency gap, mean-median difference, and partisan bias) within a given state when viewed in isolation. All three of these metrics yield the same results across both elections. All three metrics identify PLANC2333 as an outlier. Their conclusions differ only across states, while PLANC2333 is identified as an outlier, S.865 is not. On the other hand, AUS identifies S.865 as an outlier on state-mean level, while the partisan symmetry metrics did not. The same comparison does not identify Texas' PLANC2333 as an outlier unless the analysis is conducted at a finer spatial scale, such as the district level.

The difference in results across metrics and the identification of outliers are in line with what these metrics were designed to capture. The three partisan symmetry metrics that evaluate disproportionality between votes and seats, rely exclusively on aggregated electoral outcomes and abstract from the spatial distribution of voters within and across districts (DeFord et al., 2023). As such, they identify Texas' PLANC2333 as an outlier because it exhibits partisan asymmetry, but they do not suggest that South Carolina's S.865 is an outlier. S.865 is not flagged because it represents a case of ethnolinguistic gerrymandering where the cracking and packing of ethnic and language minority voters does not manifest in partisan asymmetry (Lublin, 1997). The GEO metric incorporates spatial constraints by evaluating whether alternative district configurations could plausibly yield additional seats for a party, yet it remains outcome-oriented and does not directly measure the underlying demographic configuration of the electorate (Ratliff et al., 2025). Similarly, it identifies PLANC2333 as an outlier, but not S.865. In contrast, AUS is designed to quantify the spatial distribution of ethnic and language minority voters across districts, independent of electoral results and thus partisan symmetry (Saporito & Maliniak, 2022). Therefore, AUS is better suited to identify ethnolinguistic bias in redistricting, but its interpretation is still sensitive to the choice of scale in aggregation (Saporito & Maliniak, 2022; Lee et al., 2025).

The scale-dependent difference in results between metrics reflects MAUP and its influence on the suitability of gerrymandering metrics (Openshaw, 1984; Lee et al., 2025). This difference in the identification of outliers between metrics highlights that the suitability of AUS for providing quantitative evidence of ethnolinguistic gerrymandering depends on the spatial scale of aggregation (Saporito & Maliniak, 2022). For example, Figure 22 shows that Texas' PLANC2333 has an AUS value of -1.326, which falls within the distribution of the simulated ensemble. This reflects that the value is an average of the district-level AUS values for the state. As Texas has 38 districts, the districts with extreme values are masked by the average when aggregated at the state level. This means, when aggregated, the opposing extremes offset one another, masking extreme district-level values and reducing their influence on the state-level average (Griffith, 2024). By contrast, South Carolina's smaller number of districts limits this masking effect, making extreme segregation values more influential in state-level AUS estimates. As a result, state-level AUS values tend to underestimate the degree of ethnolinguistic gerrymandering in states with many districts, even when AUS values are extremes in specific districts. This limitation does not

compromise the suitability of AUS as an ethnolinguistic gerrymandering metric; rather, it highlights the importance of MAUP when aggregation is performed.

The results, when viewed through the lens of MAUP, show that the five metrics considered in this study differ fundamentally in their analytical focus and their suitability for quantifying ethnolinguistic gerrymandering. First, the partisan symmetry metrics (efficiency gap, mean-median difference, and partisan bias) are designed to identify cases of gerrymandering with disproportionality between votes and seats. Therefore, these metrics perform well when gerrymandering manifests in partisan asymmetry, as in Texas' PLAN2333, but they are not suited to identify cases of ethnolinguistic gerrymandering where the packing and cracking of ethnic and language minority voters does not translate into partisan asymmetry (DeFord et al., 2023; Zhu, 2024). As a result, the partisan symmetry metrics do not identify South Carolina's S.865 as an outlier.

Second, the GEO metric incorporates geographic constraints by evaluating whether alternative district configurations could create competitive districts, making it more sensitive to boundary changes than the partisan symmetry measures, which consider only vote share aggregates. However, GEO remains fundamentally a partisan gerrymandering metric, and in this case, it does not provide substantially different insights from partisan symmetry metrics when used alone (Ratliff et al., 2025). Still, since GEO considers alternative districting configurations, it can serve as a supporting metric in contexts where MAUP limits the interpretability of other metrics (Lee et al., 2025).

Third, AUS is the only metric used in this study that is designed to quantify ethnolinguistic packing and cracking as a spatial phenomenon. The metric does not consider electoral outcomes and thus provides information that partisan symmetry GEO metrics cannot capture (Saporito & Maliniak, 2022). As such, AUS is an important addition for identifying ethnolinguistic gerrymandering. However, because AUS is sensitive to scale-dependent aggregation effects associated with MAUP, it should not be used in isolation to quantify the electoral consequences of ethnolinguistic gerrymandering.

5.1.2 Outlier Analysis and the Quantification of Ethnolinguistic Gerrymandering

The results of this study indicate that quantifying ethnolinguistic gerrymandering should not be done using a single metric, including AUS, due to MAUP. While AUS is designed to reflect the packing and cracking of ethnic and language minority voters, the case of Texas demonstrates that MAUP can mask extreme values when the states consist of many districts. This finding aligns with Ramachandran and Gold (2018), who argue that individual gerrymandering metrics are inherently sensitive to MAUP because aggregation is an inescapable feature of the redistricting process and its quantification. This makes single metrics unsuitable for inference of discriminatory intent, even if they can quantify the packing and cracking of ethnic and language minority voters (Lee et al., 2025).

Heuristic simulation-based approaches, such as the outlier-analysis approach used in this study, address limitations for inference of intent by comparing enacted redistricting plans to a simulated ensemble of valid alternatives that serve as a neutral benchmark against which outliers can be identified. Chen and Rodden (2015) show that ensemble comparisons allow courts and researchers to separate asymmetries attributable to natural political geography or compliance with redistricting criteria from those that cannot be plausibly justified on neutral grounds. Building on this logic, an outlier-based approach enables the joint evaluation of complementary metrics, AUS and GEO in this study, by identifying whether the calculated values are outliers relative to the neutral benchmark. As such, outlier analysis enables the integration of multiple

metrics to provide quantitative evidence for the inference of intent by combining their strengths while accounting for MAUP.

This limitation can be addressed by using AUS together with at least one of the four other metrics as done in the case of Texas' PLANC2333. For example, while AUS identifies packing and cracking of ethnic and language minority voters, the GEO metric provides insight into whether the patterns of cracking translate into advantages for either political party in seat outcomes. Even if the large number of seats obscures extreme district-level MAUP values, the advantages of packing and cracking are directly reflected in the GEO metric. The use of multiple metrics should not be seen as a limitation, but rather a consequence of the fact that gerrymandering operates through multiple mechanisms which are differently quantified (Barton and Eguia 2024). Ratliff et al. (2025) caution against treating any single metric as a sufficient diagnostic even within ensemble-based analyses. When used jointly, AUS and GEO address the inferential challenge caused by MAUP and highlight that no single metric can reliably identify ethnolinguistic bias in isolation. Instead, inferring intent to discriminate against ethnic and language minority voters from quantitative data will likely yield better results when at least two distinct metrics are used.

5.1.3 GEO, AUS, and Outlier Analysis in the Context of *Alexander v. South Carolina State Conference of the NAACP*

The use of two distinct metrics to address MAUP also addresses the evidentiary standard established in *Alexander v. South Carolina State Conference of the NAACP* (2024), which requires disentangling partisan bias from the intent to discriminate on the basis of the ethnicity or language background of minority voters. Furthermore, this combination of metrics in an outlier analysis approach is consistent with prior judicial reasoning that evaluates enacted plans relative to neutral baselines rather than absolute fairness thresholds (Grofman, 2018; Grofman & Cervas, 2018). In this context, the GEO metric can support inference of partisan bias by quantifying how advantageous a plan is for a specific political party (Campisi et al., 2022), whereas AUS supports inference of ethnolinguistic bias without distinguishing among political parties (Saporito & Maliniak, 2022). Neither metric is sufficient on its own to support the inference of discriminatory intent (Ratliff et al., 2025). However, their combined use allows for the identification of spatial patterns reflecting ethnolinguistic packing and cracking. Moreover, when combined, they help assess whether these patterns translate into systematic representational disadvantages. By using two distinct metrics instead of a single metric that attempts to do both, partisan bias and ethnolinguistic bias remain disentangled. However, by using two metrics in parallel, the outlier analysis approach still supports the inference of intent to discriminate based on ethnicity and language backgrounds of the voters.

5.2 Limitations of the Study

MAUP and the inference of discriminatory intent may be mitigated by using two distinct gerrymandering metrics. However, MAUP remains inherent to all cases of aggregation and the method and input data used in this study is already a product of aggregation. For example, the GerryChain package used to generate the neutral simulated ensemble uses the scale of voting tabulation districts (VTDs) as its base spatial unit where demographic data of census blocks and precinct-level voting outcomes are aggregated for later use in the generation of redistricting scenarios. As noted by Lee et al. (2025), statistics outcomes vary when the same underlying data are aggregated to different spatial units, even if precautions are taken to address MAUP when studying gerrymandering. While VTDs are the most valid choice of scale for gerrymandering, since real districts are also drawn using VTDs, aggregation on a different scale such as precincts could lead to different outcomes which were not tested within the study. Due to the limitations of scale and time of this study, the effect of the coarseness of the base spatial unit was not included in the sensitivity analysis and thus its effect on the results is unknown.

While outlier analysis provides a method for quantifying gerrymandering while being aware of MAUP, it introduces its own limitations. Most importantly, the simulated ensemble is assumed to be a neutral benchmark against which plans S.865 and PLANC2333 can be evaluated, but it not definitively proven to be neutral. As noted by Chen and Rodden (2015), ensemble-based approaches that rely on a neutral benchmark are inherently conditional on such modeling choices and do not truly represent neutrality by their own virtue. Instead, the neutrality of the ensemble depends on the choice of constraints, including population equality thresholds, contiguity and compactness requirements, and the fixed number of districts, as well as on the state's geographic shape and the real spatial distribution of the ethnic and language minority voters (Duchin & Tenner, 2018). Although the sensitivity analysis conducted for the research areas of this study indicated that the fixed parameters are not impactful, this may not be generalized to other states, such as Florida, which has a different geographic configuration and demographic patterns (Haas et al., 2020).

In combination, these limitations reflect the limits of quantitative analysis as a method for addressing the evidentiary demands set forth in *Alexander v. South Carolina State Conference of the NAACP* (2024). MAUP cannot be entirely mitigated; however, it can be accounted for by using two gerrymandering metrics at a minimum while applying outlier analysis to reduce the potential influence of individual district boundaries, the zoning problem, on the results of the study. Still, quantitative gerrymandering metrics, even when evaluated against a neutral benchmark, should not be treated as definitive evidence of gerrymandering (Grofman & King, 2007; Cervas & Grofman, 2020). The quantitative insights provided in this study should be seen as complementary to other quantitative and qualitative approaches to judicial challenges to gerrymandering.

5.3 Recommendations for Future Research

When considered in the context of *Alexander v. South Carolina State Conference of the NAACP* (2024), the study limitations offer several avenues for future research. First, future research could expand the analysis to states other than South Carolina and Texas, preferably with the coverage of the entire U.S. Expanding the coverage would be a particularly interesting direction of research as the comparison of South Carolina and Texas already yielded that sensitivity of the findings to the scale problem of MAUP varies by the statutory number of districts and the geographic shape of the state. Expanding the scope to cover all U.S. states would enable researchers to evaluate how the scale problem of MAUP differs by state and thus refine the use of gerrymandering metrics to reflect state-specific conditions better.

Second, the number of identified ethnolinguistic gerrymandering metrics is extremely limited compared to the family of partisan symmetry scores such as the efficiency gap. As stated in Chapter 1. Introduction, partisan symmetry scores no longer meet the evidentiary standards set in *Alexander v. South Carolina State Conference of the NAACP* (2024) on their own. Expanding the number of ethnolinguistic metrics by creating alternatives to AUS would be academically and judicially relevant research. For example, future research could broaden the scope of ethnolinguistic gerrymandering metrics by incorporating measures of voter displacement and representational discontinuity, which have been shown to disproportionately affect minority voters following redistricting (Niven et al., 2022; Niven & Solimine, 2022). A larger number of ethnolinguistic metrics would strengthen quantitative analysis approaches to study the influence of gerrymandering on the electoral representation of ethnic and language minority voter blocs and thus contribute quantitative insights to judicial challenges to gerrymandering.

Third, integrated approaches that combine partisan and ethnolinguistic bias metrics should be developed. The comparison of ethnolinguistic measures like AUS and spatial partisan metrics like GEO in this study demonstrates that these measures provide complementary information and offer an advantage over approaches that use a single metric. Prior work has similarly emphasized that gerrymandering is a multi-dimensional phenomenon for which no single metric can provide

a complete picture, and that multiple measures are more informative than relying on an individual score (Cervas & Grofman, 2020; Barton & Eguia, 2024; Ratliff et al., 2025). Evaluating ethnolinguistic and partisan biases with spatial metrics would allow researchers to distinguish more effectively between ethnolinguistic packing and cracking compared to partisan asymmetry alone. Future research should formalize such a framework to support more robust analysis of gerrymandering's influence on the representation of ethnic and language minority voters and districting outcomes while following the recommendation of Ratliff et al. (2025) to not rely on individual metrics.

Moreover, two adjacent lines of research offer opportunities to extend and apply the outlier analysis approach to the study of ethnolinguistic gerrymandering. For example, future research could study integrate newly developed ethnolinguistic gerrymandering metrics with causal analyses of voter turnout to examine whether packing and cracking translate into reduced political participation (Fraga, 2016). Additionally, future research could extend ensemble-based approaches such as outlier analysis to evaluate how ethnolinguistic gerrymandering influences broader electoral dynamics, such as competitiveness and incumbency advantage, building on the foundations set by the simulation-based frameworks for quantifying electoral competition (Cottrell, 2019).

6. Conclusion

The objective of this study was to examine the role of spatial patterns of gerrymandering in supporting the inference of ethnolinguistic bias under the constraints imposed by MAUP. The findings indicate that spatial patterns, when evaluated quantitatively, can support the inference of ethnolinguistic bias, but only when interpreted relative to neutral benchmarks and with multiple metrics used in parallel. On their own, spatial patterns are insufficient to support inference.

The review and comparison of existing assessment methods demonstrate that commonly used gerrymandering metrics vary substantially in how they represent the spatial patterns of gerrymandering. AUS quantifies the patterns of packing and cracking of ethnic and language minority voters while remaining susceptible to MAUP. The GEO metric addresses MAUP but cannot be used on its own to identify ethnolinguistic bias. When MAUP is considered, it is clear that individual metrics are limited in their interpretability and should be used in parallel.

In conclusion, this study demonstrates that quantitative analysis of spatial patterns can support the inference of ethnolinguistic bias in redistricting, but it should not be used as definitive evidence. The study's contribution lies in clarifying the conditions under which inference of ethnolinguistic bias is strengthened and the limitations that remain unavoidable due to MAUP. As such, the findings should be understood as one component of a broader quantitative approach of inferring ethnolinguistic bias, which should be complemented by additional quantitative tests, qualitative analysis, and legal interpretation in future research.

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Annex A – Spatial Data

Table 5. Overview of collected datasets.

Reference	Date accessed	Data type	Coordinate system	Spatial extent	Temporal extent (years)
Manson, S., Schroeder, J., Van Riper, D., & Ruggles, S. (2024). <i>IPUMS National Historical Geographic Information System: Version 19.0</i> [Data set]. Minneapolis, MN: IPUMS.	11/08/2025	Tabular	NA	South Carolina & Texas	2020
Manson, S., Schroeder, J., Van Riper, D., & Ruggles, S. (2024). <i>IPUMS National Historical Geographic Information System (NHGIS): 2020 Census Block Boundary Files</i> [Shapefile]. Minneapolis, MN: IPUMS.	11/08/2025	Vector	NAD83 (EPSG:4269)	South Carolina & Texas	2020
U.S. Census Bureau. (2020). <i>2020 TIGER/Line Shapefiles: Voting Districts (VTDs)</i> . Washington, DC: U.S. Department of Commerce. Retrieved from https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.2020.html#list-tab-790442341	11/08/2025	Vector	Conus Albers (EPSG:5070)	U.S.	2020
Redistricting Data Hub. (2021). <i>2020 Texas precinct boundaries and election results</i> [Data set]. Retrieved from https://redistrictingdatahub.org/dataset/2020-tx-general-precinct-election-results/	19/09/2025	Vector	NAD83 (EPSG:4269)	Texas	2020
Redistricting Data Hub. (2021). Congressional district boundaries for Texas, 2012–2021. Retrieved January 20, 2021	19/09/2025	Vector	NAD83 (EPSG:4269)	Texas	2012-2021
Redistricting Data Hub. (2022). <i>South Carolina 2022 general election precinct boundaries and election results</i> . Retrieved from https://github.com/openelections/openelections-data-sc	11/08/2025	Vector	NAD83 (EPSG:4269)	South Carolina	2022
Redistricting Data Hub. (2022). <i>2022 South Carolina congressional districts approved plan (S.865)</i> [Shapefile]. Retrieved from	11/08/2025	Vector	NAD83 (EPSG:4269)	South Carolina	2022

https://redistrictingdatahub.org/dataset/2022-south-carolina-congressional-districts-approved-plan/					
Redistricting Data Hub. (2025). <i>2025 Texas congressional districts enacted plan (PLANC2333)</i> [Shapefile]. Retrieved from https://redistrictingdatahub.org/dataset/2025-texas-congressional-districts-approved-plan/	19/09/2025	Vector	NAD83 (EPSG:4269)	Texas	2025
U.S. Census Bureau. (2020). <i>2020 TIGER/Line Shapefiles: Congressional Districts (116th Congress)</i> . Washington, DC: U.S. Department of Commerce. Retrieved from https://catalog.data.gov/dataset/tiger-line-shapefile-2020-nation-u-s-116th-congressional-districts	11/08/2025	Vector	Conus Albers (EPSG:5070)	U.S.	2020
U.S. Census Bureau. (2020). <i>2020 TIGER/Line Shapefiles: Voting Districts (VTDs)</i> . Washington, DC: U.S. Department of Commerce. Retrieved from https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.2020.html#list-tab-790442341	11/08/2025	Vector	Conus Albers (EPSG:5070)	U.S.	2020

Annex B – Manuscripts Retrieved for the Systematic Literature Review

ID	Reference	Source for Retrieval	Type of Bias Discussed	Metrics Discussed
1	Barton, Jeffrey T. (2022). "Fairness in Plurality Systems with Implications for Detecting Partisan Gerrymandering." <i>Mathematical Social Sciences</i> 117:69–90. doi:10.1016/j.mathsocsci.2022.03.002.	WoS	Partisan	Efficiency gap, Partisan bias, Mean-median difference, Declination, Responsiveness
2	Barton, Jeffrey T., and Jon X. Eguia. (2024). "A Decomposition of Partisan Advantage in Electoral District Maps." <i>Electoral Studies</i> 92:102871. doi:10.1016/j.electstud.2024.102871.	WoS	Partisan	Efficiency gap
3	Bowen, Daniel C. 2014. "Boundaries, Redistricting Criteria, and Representation in the U.S. House of Representatives." <i>American Politics Research</i> 42(5):856–95. doi:10.1177/1532673X13519127.	WoS	Racial	Polspy-Popper compactness score, Reock compactness score, Coterminosity
4	Buchanan, Austin, Soraya Ezazipour, and Maral Shahmizad. (2025). "A Widespread Belief About County Splits in Political Districting Plans Is Wrong." <i>Election Law Journal: Rules, Politics, and Policy</i> elj.2024.0040. doi:10.1089/elj.2024.0040.	WoS	Partisan	County splits
5	Campisi, Marion, Thomas Ratliff, Stephanie Somersille, and Ellen Veomett. (2022). "Geography and Election Outcome Metric: An Introduction."	Snowball	Partisan	GEO Metric
6	Chatterjee, Tanima, Bhaskar DasGupta, Laura Palmieri, Zainab Al-Qurashi, and Anastasios Sidiropoulos. (2020). "On Theoretical and Empirical Algorithmic Analysis of the Efficiency Gap Measure in Partisan Gerrymandering." <i>Journal of Combinatorial Optimization</i> 40(2):512–46. doi:10.1007/s10878-020-00589-x.	WoS	Partisan	Efficiency gap

7	Cottrell, David. (2019). "Using Computer Simulations to Measure the Effect of Gerrymandering on Electoral Competition in the U.S. Congress." <i>Legislative Studies Quarterly</i> 44(3):487–514. doi:10.1111/lsq.12234.	WoS	Partisan	
8	DeFord, Daryl, Natasha Dhamankar, Moon Duchin, Varun Gupta, Mackenzie McPike, Gabe Schoenbach, and Ki Wan Sim. (2023). "Implementing Partisan Symmetry: Problems and Paradoxes." <i>Political Analysis</i> 31(3):305–24. doi:10.1017/pan.2021.49.	Snowball	Partisan	Mean-median difference, Partisan bias, Partisan Gini
9	Dobbs, Kiera W., Rahul Swamy, Douglas M. King, Ian G. Ludden, and Sheldon H. Jacobson. (2024). "An Optimization Case Study in Analyzing Missouri Redistricting." <i>INFORMS Journal on Applied Analytics</i> 54(2):162–87. doi:10.1287/inte.2022.0037.	WoS	Partisan	Seats-votes curve, Partisan bias, Polsby-Popper compactness score, Perimeter-based compactness measure, County splits
10	Dube, M. P. (2022). Graphical metrics for analyzing district maps. <i>Journal of Computational Social Science</i> .	WoS	Partisan	Edge-cut growth, Excess edge, Internal boundary growth, Polsby-Popper compactness score, Reock compactness score, Convex hull compactness score
11	Eguia, J. X. (2022). A Measure of Partisan Advantage in Redistricting. <i>Election Law Journal: Rules, Politics, and Policy</i> , 21(1), 84–103. https://doi.org/10.1089/elj.2020.0691	WoS	Partisan	Jurisdictional partisan advantage
12	Fraga, B. L. (2016). <i>Redistricting and the Causal Impact of Race on Voter Turnout</i> .	WoS	Racial	NA
13	Goedert, N., Hildebrand, R., Pierson, M., Travis, L., & Fravel, J. (2024). <i>Black Representation and District Compactness in Southern Congressional Districts</i> .	WoS	Racial	Polsby-Popper compactness score, Reock compactness score

14	Griffith, D. A. (2024). Spatial Autocorrelation and Political Redistricting: A Task for the Uniform Distribution.	WoS	Partisan	NA
15	Haas, C., Hachadoorian, L., Kimbrough, S. O., Miller, P., & Murphy, F. (2020). Seed-Fill-Shift-Repair: A redistricting heuristic for civic deliberation. PLOS ONE, 15(9), e0237935. https://doi.org/10.1371/journal.pone.0237935	WoS	Partisan	Population deviation, Polsby-Popper compactness score, Reock compactness score, County splits
16	Nagle, J. F., & Ramsay, A. (2021). On Measuring Two-Party Partisan Bias in Unbalanced States. Election Law Journal: Rules, Politics, and Policy, 20(1), 116–138. https://doi.org/10.1089/elj.2020.0674	WoS	Partisan	Partisan bias
17	Niven, D., Combs, B. H., Norwood, C., Rossiter, K. E., & Solimine, M. E. (2022). The boundaries of confusion: Gerrymandering and racial disparities in state House and congressional district line congruity. Social Science Quarterly, 103(6), 1507–1518. https://doi.org/10.1111/ssqu.13213	WoS	Racial	Unnecessary district overlaps
18	Niven, D., & Solimine, M. E. (2022). Representing People and Places: Castaway Voters and the Racial Disparity in Redistricting. Election Law Journal: Rules, Politics, and Policy, 21(2), 171–186. https://doi.org/10.1089/elj.2021.0032	WoS	Racial	Racial disparity
19	Openshaw, S. (1984). The Modifiable Areal Unit Problem. Geo Books. https://www.uio.no/studier/emner/sv/iss/SGO9010/openshaw1983.pdf	Core theoretical manuscript	NA	NA
20	Plane, D. A., Tong, D., & Lei, T. (2019). Inter-person Separation: A New Objective Standard for Evaluating the Spatial Fairness of Political Redistricting Plans. Geographical Analysis, 51(3), 251–279. https://doi.org/10.1111/gean.12175	WoS	Partisan	Inter-person separation
21	Powell, R. J., Clark, J. T., & Dube, M. P. (2020). Partisan Gerrymandering, Clustering, or Both? A New Approach to a Persistent Question. Election Law Journal: Rules, Politics, and Policy, 19(1), 79–100. https://doi.org/10.1089/elj.2019.0605	WoS	Partisan	Seats-change measure, Sullivan diversity index

22	Ramachandran, G., & Gold, D. (2018). Using Outlier Analysis to Detect Partisan Gerrymanders: A Survey of Current Approaches and Future Directions. <i>Election Law Journal: Rules, Politics, and Policy</i> , 17(4), 286–301. https://doi.org/10.1089/elj.2018.0503	Snowball	Partisan	Efficiency gap, Mean-median difference, Partisan bias
23	Sánchez, F. (2011). Racial Gerrymandering and Geographic Information Systems: Subverting the 2011 Texas District Map with Election Technologies.	WoS	Racial	NA
24	Saporito, S., & Maliniak, D. (2022). Using the areal unit segregation measure to identify racially “packed” and “cracked” legislative districts. <i>Electoral Studies</i> , 80, 102526. https://doi.org/10.1016/j.electstud.2022.102526	Snowball	Racial	Areal unit segregation measure
25	Sun, S. (2021). Developing a Comprehensive and Coherent Shape Compactness Metric for Gerrymandering. <i>Annals of the American Association of Geographers</i> , 111(1), 175–195. https://doi.org/10.1080/24694452.2020.1760779	WoS	Partisan	Cumulative circle coverage metric
26	Swamy, R., King, D. M., & Jacobson, S. H. (2025). Multiobjective Optimization for Politically Fair Districting: A Scalable Multilevel Approach.	WoS	Partisan	Partisan bias, Competitiveness, Polsby-Popper compactness score, Reock compactness score
27	Swamy, R., King, D. M., Ludden, I. G., Dobbs, K. W., & Jacobson, S. H. (2024). A practical optimization framework for political redistricting: A case study in Arizona. <i>Socio-Economic Planning Sciences</i> , 92, 101836. https://doi.org/10.1016/j.seps.2024.101836	WoS	Partisan	Competitiveness, Perimeter-based compactness, Population equality, VRA score
28	Vargas, R., Cano, C., Del Toro, P., & Fenaughty, B. (2023). The Racial and Economic Foundations of Municipal Redistricting. <i>Social Problems</i> , 70(4), 935–960. https://doi.org/10.1093/socpro/spab076	WoS	Racial	NA
29	Wang, S. S.-H. (2016). Three tests for practical evaluation of partisan gerrymandering. <i>Stanford Law Review</i> , 68(6), 1263–1321.	WoS	Partisan	Mean-median difference

30	Zhu, Z. (2024). Partisan Gerrymandering and Political Geography: A Simulation-Based Analysis on the Effectiveness of the Efficiency Gap and Mean–Median as Measures of Partisan Gerrymandering.	WoS	Partisan	Efficiency gap, mean-median difference
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